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An Empirical Study of English Pronunciation Assessment Using Goodness of Pronunciation Scoring

by

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Abstract

Goodness of Pronunciation (GOP) is a central feature in computer-assisted pronunciation training, yet the literature treats GOP as a single scalar while implementations differ in posterior geometry, acoustic model, alignment, and scorer. Reported improvements are therefore difficult to attribute to any one design choice. This thesis examines how GOP validity depends on representation, acoustic-model posteriors, and scoring architecture under a reproducible protocol on Speechocean762, a public non-native English corpus with expert phoneme scores.

A controlled factorial design varies one independent factor at a time while holding the official speaker-disjoint partition and a consistent phoneme-level evaluation protocol fixed. Confirmatory hypotheses are evaluated with bootstrap confidence intervals and paired correlation contrasts. Behavioral heterogeneity and failure modes are examined exploratorily.

Traditional Kaldi canonical log phone posterior scoring is positively associated with expert ratings (PCC $= 0.323$). Competitor-aware GOP representations outperform scalar LPP, and alternative self-supervised posteriors and alignment-free decoding yield further gains. The strongest non-learned configuration reaches PCC $= 0.498$; learned Transformer scoring on frozen confusion profiles achieves PCC $= 0.671$. Learned scoring extends the confirmatory GOP questions rather than forming a separate research question.

GOP behavior varies across phonemes, speakers, and sentence-score tertiles, so pooled

correlation is incomplete. Error analysis distinguishes five failure modes: Alignment Shift, Phonetic Confusion, L2 Accent Distortion, Contextual Coarticulation, and Speaker Outlier. The study contributes a reproducible factorial benchmark and evidence that GOP is a structured feature family rather than a single scalar. The findings apply to read speech from Mandarin-L1 learners in Speechocean762 and require cross-corpus validation.

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Contents

Abstract **i**

Acknowledgments **iii**

1 Introduction **1**

1.1 Background 1

1.2 Problem Statement 2

1.3 Research Gap 3

1.4 Research Objectives 3

1.5 Research Questions and Hypotheses 4

1.6 Scope of the Study 5

1.7 Research Contributions 6

1.8 Thesis Organization 6

2 Background and Related Work **7**

2.1 Automatic Pronunciation Assessment 7

2.2 English Pronunciation Assessment 8

2.3 Goodness of Pronunciation 8

2.4 GOP Formulations 9

2.5 Acoustic Models 10

2.6 Learned GOP-Based Scoring 11

2.7 Pronunciation Assessment Datasets 13

2.8 Evaluation Metrics 13

2.9 Related Empirical Studies 14

2.10 Literature-Based Gap 16

3 Research Methodology **18**

3.1 Research Design 18

3.2 Hypothesis Operationalization 21

3.3 Dataset 24

3.4 Data Preprocessing 26

3.5 Forced Alignment 28

3.6 Acoustic Models 31

3.7 GOP Computation 33

3.8 GOP Representations 34

3.9 Pronunciation Scoring 34

3.10 Evaluation Metrics 38

3.11 Statistical Analysis 39

3.12 Experimental Protocol 42

3.13 Reproducibility 43

4 Experimental Design and Results **45**

4.1 Experimental Setup 45

4.2 GOP Effectiveness 45

4.3 GOP Representation 47

4.4 Acoustic Model Dependency 47

4.5 Behavioral Analysis 48

4.6 Learned GOP Scoring 49

4.7 Statistical Analysis 50
4.8 Error Analysis 52
4.9 Summary of Findings 53

5 Discussion **54**

5.1 Overview of the Main Findings 54
5.2 Discussion of RQ1: Effectiveness of GOP-based Pronunciation Assessment 54
5.3 Discussion of RQ2: Effect of GOP Representation 55
5.4 Discussion of RQ3: Effect of Acoustic Model 55
5.5 Discussion of RQ4: Behavioral Patterns and Failure Modes of GOP . . . 56
5.6 Overall Discussion: What Do the Findings Tell Us About GOP? 56
5.7 Comparison with Previous Studies 57
5.8 Practical Implications 57
5.9 Threats to Validity 58
5.10 Chapter Summary 59

6 Conclusion and Future Work **60**

6.1 Summary of the Thesis 60
6.2 Key Empirical Findings 60
6.3 Research Contributions 61
6.4 Limitations of the Study 62
6.5 Directions for Future Work 62

List of Figures

3.1 Overall Research Design and Methodological Workflow 18

3.2 Kaldi CTM as the single alignment source: phone sequence *y* for alignment-free GOP, and time spans only when a formula requires a Kaldi window. . 29

6.1 Executive research progression of the thesis. 60

List of Tables

2.1 GOP-related representations used in this thesis 10

3.1 Systematic Experimental Factors and Controlled Configurations 19

3.2 Comprehensive Experimental Matrix Across Investigation Phases 20

3.3 Mapping of research questions to hypotheses, experimental phases, and analyses 23

3.4 Locked Speechocean762 evaluation set used in every experimental phase 24

3.5 Human score scales used in this study 25

3.6 Locked preprocessing steps applied before GOP computation 27

3.7 Alignment artifacts and their role under the protocol lock 30

3.8 Acoustic models used to generate phoneme posteriors 31

3.9 Locked learned-scoring hyperparameters 38

3.10 Summary of quantitative evaluation metrics 39

3.11 Locked score spaces for hypothesis tests and model contrasts 40

3.12 Planned H3 paired contrasts (direct scoring, calibrated $[0,2]$) 41

3.13 RQ4 strata, locked GOP, and minimum N for reporting PCC/SCC 41

3.14 Locked residual-taxonomy rules 42

3.15 Operating protocol lock for all investigation phases 43

4.1 Mean canonical LPP by human-score band on the official test set 46

4.2 Traditional GOP versus expert phoneme scores. PCC and SCC are computed on raw GOP. MAE and MSE use the train-only affine map clipped to $[0,2]$. Intervals are percentile 95% bootstrap intervals on the test tokens ($n_{\text{boot}} = 1{,}000$). 46

4.3 Performance comparison across GOP feature representations on the official Speechocean762 test partition ($N = 47{,}369$). All representations are extracted from the Kaldi M13 (TDNN-F Chain) acoustic model and evaluated with train-only linear estimators clipped to $[0,2]$. 47

4.4 Acoustic model performance comparison on the official Speechocean762 test set ($N = 47{,}369$). All models use direct scoring with train-only univariate affine maps clipped to $[0,2]$. Confidence intervals denote percentile 95% bootstrap intervals ($n_{\text{boot}} = 1{,}000$). 48

4.5 Planned paired ΔPCC contrasts for hypothesis H3. All intervals exclude zero. 48

4.6 Behavioral analysis of traditional GOP across phoneme classes, speaker distributions, and sentence-accuracy tertiles on the official Speechocean762 test partition ($N = 47{,}369$, 125 speakers). All strata evaluate Kaldi canonical LPP with frozen baseline calibration. 49

4.7 Performance of learned scoring architectures across acoustic features and embedding configurations on the official Speechocean762 test partition ($N = 47{,}369$). Predictions are clipped to $[0,2]$. 50

4.8 Percentile 95% bootstrap confidence intervals for Pearson correlation ($n_{\text{boot}} = 1{,}000$) on the official Speechocean762 test set ($N = 47{,}369$). All models are evaluated in calibrated $[0,2]$ score space. 51

4.9 Paired bootstrap difference tests (ΔPCC, $n_{\text{boot}} = 1{,}000$) for planned system comparisons. All confidence intervals exclude zero. 51

4.10 Multi-seed training stability across seeds $\{0,1,2,3,4\}$ on the official test set. 52

4.11 Error taxonomy distribution on the official Speechocean762 test partition ($N = 47{,}369$) comparing the Direct SSL baseline and the Neural Transformer system. 52

4.12 Mean absolute error $|\hat{y} - y|$ at score scale boundaries on the official test set. 52

4.13 Overall performance ranking across primary GOP assessment paradigms on the official Speechocean762 test set ($N = 47{,}369$). All reported configurations use calibrated $[0,2]$ score spaces. 53

Chapter 1
Introduction

1.1 Background

In an increasingly globalized world, accurate English pronunciation is a prerequisite for effective communication in academic, professional, and social settings. Computer-Assisted Pronunciation Training (CAPT) systems address this need by combining automatic speech recognition with pedagogical feedback, allowing learners to practice outside the classroom [19, 2]. Most deployed CAPT pipelines ultimately require a phoneme-level signal: a score or diagnostic label attached to each target phone, not merely a holistic word or sentence rating. Without that granularity, learners cannot tell whether an error is a substitution, a deletion, or an acceptable accent variant.

The Goodness of Pronunciation (GOP) framework, formalized by Witt and Young [16], remains the dominant approach for deriving such phoneme-level evidence from acoustic-model posteriors. GOP expresses, for each canonical phone, how well the observed speech supports that phone relative to competing hypotheses. Decades of CAPT research have built scoring, mispronunciation detection, and diagnostic tools on GOP or its variants [7, 2]. More recently, self-supervised speech encoders such as Wav2Vec 2.0 [1] and multilingual extensions such as XLSR-53 [3] have changed how posteriors are obtained, while learned scorers inspired by GOPT [7] have changed how posteriors are mapped to human-interpretable grades.

Despite this progress, the literature rarely compares GOP configurations under a single locked protocol. Acoustic model, alignment artifact, GOP representation, and scoring function are often changed simultaneously, which makes it impossible to attribute a reported gain to a specific factor. The open-source Speechocean762 corpus [19] provides expert phoneme scores, a documented L2 speaker population, and an official speaker-disjoint train/test partition; these conditions support reproducible empirical evaluation rather than one-off system demonstrations.

This thesis therefore does *not* propose a new end-user CAPT product. It conducts a controlled factorial empirical study that measures and decomposes GOP validity on Speechocean762: whether GOP correlates with expert ratings, which representations and acoustic models change that correlation, and where GOP fails across phonemes and speakers. Chapter 3 formalizes the protocol; Chapters 4–5 report and interpret the findings.

1.2 Problem Statement

In both research papers and engineering practice, the term "GOP" is used as if it denoted a single scalar quantity. In reality, GOP is an operational family: canonical Log Phone Posterior (LPP), Log Posterior Ratio (LPR) against the strongest competitor, low-dimensional GOP vectors, full LPP+LPR confusion profiles, span-averaged scores tied to forced-alignment boundaries, and alignment-free formulations that marginalize over decoding graphs [16, 2, 7]. Each variant conditions on a particular acoustic model, a particular phone inventory, and a particular temporal segmentation strategy. Reporting one Pearson correlation coefficient without stating those choices therefore answers an ill-posed question.

A second problem is confounding. A higher GOP-human correlation after switching from the Kaldi LibriSpeech TDNN-F Chain Model M13 baseline to a Wav2Vec 2.0 checkpoint [1, 3] may reflect the encoder, the CTC head, the alignment graph, a different phone mapping, or a new learned scorer, not "GOP itself." Without factorial control (holding the dataset, speaker split, evaluation unit, and inference procedure fixed while varying one factor at a time), cross-study comparisons remain descriptive rather than causal.

Third, pooled phoneme-level correlation can obscure heterogeneous behavior. GOP may track expert scores for some vowel classes and fail for others; it may be stable for one L2 speaker and unstable for another; large residuals may arise from boundary misalignment, acoustic confusion, acceptable accent, sentential context, or speaker-specific variation [19]. A single global metric does not reveal those failure modes.

The problem addressed by this thesis is therefore methodological and empirical: *how*

to measure GOP validity and its sensitivity to representation, acoustic-model posteriors, and scoring architecture under a reproducible protocol on a standard non-native English benchmark, and *how to characterize where GOP-based assessment succeeds or fails when examined by phoneme and speaker.* Product-level concerns such as user-interface design, real-time latency targets, or demographic fairness beyond the corpus under study are outside the present problem formulation.

1.3 Research Gap

Prior CAPT and pronunciation-assessment work leaves three gaps that motivate this study. First, *representational ambiguity*: publications and toolkits apply the single label "GOP" to distinct feature geometries (scalar LPP, competitive LPR, rank-2 vectors, and GOPT-style confusion profiles), so it remains unclear which geometric aspect carries the diagnostic signal [16, 7, 2]. Second, *confounded comparisons*: benchmark papers often change the acoustic model, alignment pipeline, GOP formula, and scorer jointly, preventing reported gains from being attributed to a specific design choice [1, 19]. Third, *limited diagnostic analysis*: few studies stratify correlation by phoneme class, speaker, or score band, or taxonomize large residuals using expert markup. The literature synthesis supporting each gap is developed in Section 2.10 of Chapter 2; this thesis addresses all three through a controlled factorial study on Speechocean762 under the locked protocol of Sections 1.6 and 3.1.

1.4 Research Objectives

The general aim of this thesis is to quantify and decompose the validity of GOP-based English pronunciation assessment under a reproducible experimental protocol. The following specific objectives support that aim and map directly to the research questions in Section 1.5.

1. **Establish a protocol-locked baseline.** Measure the association between traditional GOP (canonical Kaldi LPP) and expert phoneme scores on the official Speechocean762 test partition, thereby answering RQ1 and testing H1.

2. **Isolate GOP representation effects.** Compare scalar LPP, LPR, a rank-2 GOP vector, and an 84-dimensional LPP+LPR profile under the same Kaldi extract and linear scoring family, holding all other factors fixed (RQ2 / H2).

3. **Quantify acoustic-model sensitivity.** Compare GOP-human agreement when posteriors are drawn from Kaldi M13 versus off-the-shelf Wav2Vec2-XLSR-53 and Wav2Vec2-lv60 espeak CTC checkpoints, including alignment-free GOP variants for the SSL models (RQ3 / H3).

4. **Characterize behavioral patterns and failure modes.** Analyze stratified performance across phoneme classes, individual speakers, and sentence-score tertiles, and classify large residuals into interpretable error types using expert markup (RQ4, exploratory).

5. **Evaluate learned scoring extensions.** Assess whether MLP and Transformer scorers operating on frozen GOP features improve the GOP-to-human mapping relative to direct linear projection, with bootstrap confidence intervals and paired ΔPCC tests supporting confirmatory conclusions for RQ1-RQ3.

These objectives deliberately exclude product metrics such as phoneme error rate minimization, sub-second latency, or user-interface studies. The success criteria are statistical and diagnostic: measurable association with expert scores, detectable differences among locked configurations, and interpretable residual structure, not deployment of a standalone CAPT application.

1.5 Research Questions and Hypotheses

This section states the four research questions (RQ1-RQ4) and the three confirmatory hypotheses (H1-H3). RQ1-RQ3 are confirmatory; RQ4 is exploratory and has no H4. Learned GOP scoring is a modeling extension of RQ1-RQ2, not a fifth question. Operational definitions, rejection rules, and the mapping to experimental phases are given in Section 3.2.

Research Questions. RQ1. How effectively does GOP-based scoring correlate with human pronunciation ratings?

RQ2. How do different GOP representations affect English pronunciation assessment when the acoustic extract and linear scoring family are held fixed?

RQ3. To what extent does the acoustic model that produces phoneme posteriors affect GOP-based assessment?

RQ4. What are the behavioral characteristics and failure modes of GOP-based assessment across phonemes and speakers?

Hypotheses. These confirmatory questions are tested through three directional hypotheses.

H1 (RQ1). GOP scores are positively associated with human ratings: Pearson correlation between canonical Kaldi LPP and expert phoneme scores satisfies $\rho > 0$.

H2 (RQ2). GOP representations differ in agreement with human ratings: at least one locked pair (i, j) has $\Delta\text{PCC}_{i-j} \neq 0$.

H3 (RQ3). GOP-human PCC differs among Kaldi M13, Wav2Vec2-XLSR-53, and Wav2Vec2-lv60 on the same paired phones.

RQ4 has no H4. Its aim is a stratified description and an error taxonomy rather than a single-parameter test.

1.6 Scope of the Study

The empirical study is restricted to phoneme-level scoring of read-aloud English produced by Mandarin-L1 learners in Speechocean762 [19]. Headline metrics use the official speaker-disjoint train/test partition. The acoustic models in scope are Kaldi M13 and two frozen public Wav2Vec 2.0 espeak CTC checkpoints (Wav2Vec2-XLSR-53 and Wav2Vec2-lv60). GOP representations, scorers, and metrics are locked in Chapter 3 (Tables 3.4, 3.8, and 3.15).

The following topics are excluded: other acoustic models or alignment pipelines; evaluation on additional corpora; user-facing interface design or deployment metrics; suprasegmental prosody as a primary outcome; and formal proficiency labels beyond the corpus annotations.

Findings are not claimed to generalize to all L1 backgrounds, age groups, or spontaneous speech without further validation.

1.7 Research Contributions

This thesis contributes along four dimensions. Full statements, with evidence, appear in Section 6.3.

Theoretical. GOP is treated as a structured phonetic feature family rather than a single scalar to be replaced by an opaque scorer.

Methodological. A factorial protocol locks partition, alignment, metrics, and inference while varying one independent factor at a time.

Empirical. The study reports a locked Speechocean762 benchmark from Kaldi baseline GOP through alignment-free SSL GOP and learned Transformer scorers, with phoneme, speaker, and residual-taxonomy analyses.

Practical implications. The results suggest two scoring regimes (direct alignment-free GOP versus a hybrid Transformer on frozen profiles). These are implications, not a field-tested CAPT product.

1.8 Thesis Organization

Chapter 1 frames the problem, gaps, objectives, questions, scope, and contributions.

Chapter 2 reviews CAPT, GOP, acoustic models, datasets, metrics, and the literature-based gap.

Chapter 3 specifies the factorial protocol, data pipeline, GOP formulae, scorers, and inference rules.

Chapter 4 reports numerical results for RQ1-RQ4, learned scoring, and residual analysis.

Chapter 5 interprets mechanisms, compares prior work, and states implications and threats to validity.

Chapter 6 summarizes findings, contributions, limitations, and future work.

Chapter 2
Background and Related Work

2.1 Automatic Pronunciation Assessment

Definition and Core Functions. Computer-Assisted Pronunciation Training (CAPT) applies automatic speech recognition and signal processing to second-language (L2) phonology learning [19, 2]. A CAPT pipeline typically records learner speech, aligns or decodes it against a canonical transcript, derives phoneme-level evidence, and presents scores or diagnostics. Automatic Pronunciation Assessment (APA) refers specifically to the scoring and evaluation component: mapping acoustic evidence to grades that approximate human expert judgement.

Evolution of CAPT Technology. CAPT technology has evolved through four broad generations: (i) waveform or spectrogram displays that required phonetic expertise to interpret, (ii) HMM-based ASR and GOP scoring with forced alignment, enabling objective phoneme-level grades [16], (iii) deep neural acoustic models and end-to-end mispronunciation detection, and (iv) self-supervised encoders such as Wav2Vec 2.0 [1], multilingual pretraining [3], and Transformer scorers in the GOPT tradition [7], often combined with alignment-free GOP variants [2].

Progress in architecture has not always been matched by comparable evaluation protocols. Studies differ in corpus, train/test partition, alignment method, GOP definition, and reported metrics. Many papers publish a single correlation coefficient without locking speaker overlap or phoneme evaluation units [19]. That heterogeneity motivates the factorial empirical design adopted in this thesis: all comparisons run on Speechocean762 under a documented protocol lock, varying one factor at a time where confirmatory hypotheses require it.

2.2 English Pronunciation Assessment

English L2 pronunciation assessment operates over a fixed phoneme inventory, typically CMU/ARPABET symbols mapped to acoustic models and lexicons [19]. Assessments may target the word level (aggregating phone errors), the sentence level (fluency and prosody), or the phoneme level (local articulatory correctness). Phoneme-level scoring is the finest granularity at which substitution, deletion, and accent can be distinguished when expert markup is available.

Speechocean762, the corpus used throughout this thesis, annotates each phoneme with an expert score on a three-point scale: 2 = correct pronunciation; 1 = correct but heavily accented; 0 = wrong or missing [19]. Experts also provide qualitative tags (for example accent versus substitution) that support error analysis but are not used as training targets in the confirmatory phases. Sentence-level accuracy scores (0–10) are retained only to define speaker tertiles for exploratory behavioral analysis (RQ4); they are not treated as CEFR or proficiency labels.

The speaker population consists of Mandarin-L1 learners of English, including both adults and children, recorded under read-aloud conditions at 16 kHz. That demographic context explains some systematic L1 transfer patterns but does not license general claims about all age groups or L1 backgrounds. Cross-corpus studies on corpora such as L2-ARCTIC [20] remain relevant as related work rather than as primary evaluation benchmarks for the present study.

Phoneme-level APA therefore requires both a posterior-based feature (such as GOP) and a mapping from that feature to the human scale, with explicit control of alignment and acoustic-model choices; these topics are developed in the following sections.

2.3 Goodness of Pronunciation

The Goodness of Pronunciation (GOP) algorithm quantifies phoneme-level accuracy from acoustic-model posteriors [16]. Classically, the score of a target phone is a duration-normalized log posterior, often implemented as a ratio of the evidence for that phone against

competing phones or a phone loop. Scores that favor the canonical phone indicate stronger support; scores that favor competitors indicate substitution or poor fit. Modern toolkits often store log phone posteriors (LPP) averaged over an alignment span rather than explicit Gaussian likelihoods. The formulae used in this thesis are given in Sections 3.7 and 3.8.

GOP features feed three downstream uses: (i) direct mapping to human scores, (ii) threshold-based mispronunciation detection, and (iii) input to learned scorers that combine GOP geometry with contextual models [7]. Because GOP is defined at the phone level, it can localize errors that word-level ASR accuracy obscures.

Classical GOP depends on the acoustic model and on the temporal boundaries supplied by forced alignment [2]. Acoustic models trained on native speech may assign misleading posteriors to L2 articulations; misaligned spans assign frames to the wrong phone before GOP is computed. Self-supervised encoders change robustness, phone inventories, and alignment-free formulas [1, 3, 2]. These dependencies motivate treating GOP as a family of representations rather than a single metric (Section 2.4).

2.4 GOP Formulations

Publications use "GOP" for several distinct feature constructions [2, 7, 16]. Mathematical definitions used in this thesis appear in Sections 3.7 and 3.8. This section states the conceptual distinctions that motivate RQ2.

Span-Based Kaldi Features. Given a canonical phone p and frames aligned to its interval, *canonical LPP* is the log posterior of p averaged over the span. *Standard GOP* on the Kaldi extract used here coincides with that canonical entry. *Log Posterior Ratio (LPR)* compares the target phone with its strongest competitor: high LPR indicates that p dominates competing phones; low or negative LPR indicates confusion. A *rank-2 GOP vector* stores the canonical posterior and the strongest competitor separately. An *84-dimensional profile* concatenates all scored LPP and LPR slots, encoding local confusion geometry (the GOPT-style profile; Section 2.6).

Alignment-Free CTC GOP. Turning to CTC posteriors, span boundaries from a separate aligner may disagree with the model's peak activations. Cao et al. [2] define alignment-free GOP-S (substitution) and GOP-SD (substitution-deletion) scores on a canonical phone sequence without committing to a single frame partition. These formulations are used for the two Wav2Vec 2.0 checkpoints in RQ3.

Alignment Artifacts. Phone-level GOP traditionally assumes known phone identity and frame alignment. Kaldi phone CTMs are a common artifact and may drift on non-native speech [16, 2]. Alignment-free GOP reduces that dependence for SSL models. This thesis locks one Kaldi CTM for the Kaldi baseline; the empirical question is how GOP-human agreement changes when posteriors and graphs change, not which aligner is universally superior.

Table 2.1 lists the representations used later under the locked protocol.

Table 2.1: GOP-related representations used in this thesis

Name	Typical source	Role in study
Canonical LPP / standard GOP	Kaldi span average	RQ1 baseline; RQ2 scalar
LPR	Kaldi span	RQ2 scalar
Rank-2 vector	Kaldi span	RQ2 low-dimensional
84-d / 78-d profile	Kaldi / SSL CTC	RQ2 high-dimensional; learned scoring input
GOP-S / GOP-SD	SSL alignment-free	RQ3 SSL posterior comparison

2.5 Acoustic Models

GOP is a functional of phoneme posteriors $P(p \mid x_t)$. Changing the acoustic model changes those posteriors and therefore GOP, even when the GOP formula is written identically [16, 1]. This section describes the three posterior sources compared under RQ3; none are fine-tuned on Speechocean762 in this study.

Kaldi LibriSpeech TDNN-F Chain Model M13. The baseline system is the pretrained Kaldi Librispeech ASR Chain 1d model (M13), a factored time-delay neural network (TDNN-F) trained with the chain objective on 960 hours of Librispeech. Senone-level

outputs are mapped to phone-level log posteriors (LPP) through the recipe's senone-to-phone aggregation. Canonical LPP averaged over Kaldi phone intervals supplies the traditional GOP baseline for RQ1 and the fixed extract for RQ2 representation comparisons. The Kaldi GOP recipe for Speechocean762 builds on the same model, so the baseline can be compared with published GOP results on this corpus.

Wav2Vec 2.0 and CTC Phoneme Heads. Wav2Vec 2.0 [1] learns speech representations from unlabeled audio via a convolutional feature encoder and a Transformer context network, pretrained with a contrastive objective over quantized latent targets. For pronunciation assessment, publicly released checkpoints attach a Connectionist Temporal Classification (CTC) phoneme head fine-tuned with an espeak-derived inventory. Two checkpoints are in scope: (i) **Wav2Vec2-XLSR-53** (multilingual pretraining) [3] and (ii) **Wav2Vec2-lv60** (large English pretraining) [1]. Both are used *off-the-shelf*: weights are frozen, and no additional training or fine-tuning is performed on Speechocean762. Frame-level softmax outputs feed alignment-free GOP-S and GOP-SD computations [2]. CMU phones are mapped to espeak IPA symbols through a fixed one-to-one table locked for the entire study.

These three sources are not matched architectural competitors: Kaldi M13 and SSL-CTC systems differ in training data, phone inventory, and decoding graph. RQ3 therefore measures *sensitivity of GOP-based assessment to the posterior source*, not a claim that one toolkit universally dominates. Follow-up contrasts that freeze an SSL encoder and change only the graph (substitution versus substitution-deletion) separate decoding-graph effects from encoder effects [2]. The checkpoints remain frozen so that reported contrasts reflect posterior source and GOP formulation.

2.6 Learned GOP-Based Scoring

Linear projection from a scalar GOP feature to the human $[0,2]$ scale is the classical baseline. Recent work shows that richer posterior geometry and contextual modeling improve agreement with expert ratings [7, 10, 13].

GOPT and GOPT-Style Profiles. GOPT (Goodness-of-Pronunciation Transformer) [7] is a multi-aspect, multi-granularity architecture that consumes high-dimensional LPP+LPR profiles together with phone identity and Transformer layers to predict word- and utterance-level scores in addition to phone-level accuracy. It represents the state of the art in learned pronunciation assessment on public benchmarks.

Throughout this thesis, *GOPT-style profile* denotes the frozen LPP+LPR confusion vector only (84 dimensions on Kaldi; 78 on SSL extracts). *Full GOPT* denotes Gong et al.'s complete multi-task stack: profile plus phone embedding, Transformer encoder, and word/utterance heads. This study adopts GOPT-style profiles but does not evaluate full GOPT as a bundled system. Phone embeddings and Transformer scorers are tested as modular extensions of frozen GOP features, isolating the contribution of the scoring architecture from that of the underlying posteriors (see also Section 4.6 in Chapter 4).

The learned scoring layer includes (i) a **direct map**, a train-only affine transform from a scalar GOP feature to $[0, 2]$, (ii) **OLS**, multivariate linear regression on rank-2 vectors or full profiles, (iii) an **MLP**, a shallow nonlinear mapping on profile features, and (iv) a **Transformer**, a sequence model over phones within an utterance, optionally augmented with canonical phone embeddings. These scorers extend RQ1 (validity of mapping GOP to human scores) and RQ2 (representation under a nonlinear scorer) but do not introduce separate confirmatory hypotheses. Multi-seed training on representative configurations checks optimization stability.

Distinct from this factorial GOP comparison, joint models that simultaneously perform pronunciation assessment and mispronunciation detection share phoneme-level representations [13]. They illustrate an alternative research line (multi-task learning on CTC posteriors). The present study keeps assessment and error taxonomy analytically separate: confirmatory metrics are GOP-human correlations; qualitative markup informs exploratory failure-mode analysis (RQ4).

2.7 Pronunciation Assessment Datasets

Speechocean762. Speechocean762 is an open-source non-native English corpus for pronunciation assessment [19]. It contains 5,000 read-aloud utterances from 250 Mandarin-L1 speakers, with expert phoneme, word, and sentence scores and an official speaker-disjoint train/test partition. Public splits, phoneme-level scores on a 0-2 scale, and expert markup make it a suitable locked benchmark for this thesis. Token counts and the protocol lock are stated in Section 3.3.

Related Corpora. Relative to this locked Speechocean762 protocol, L2-ARCTIC [20] provides multi-L1 non-native English recordings but lacks the same expert phoneme scoring protocol. Other corpora appear in the literature [7, 5] and remain outside the locked protocol. Cross-corpus generalization is treated as a limitation in Chapter 6.

2.8 Evaluation Metrics

Phoneme-level pronunciation assessment requires metrics that compare model outputs with expert scores on the same $[0, 2]$ scale [19].

The agreement suite used throughout this thesis comprises four metrics: (i) **Pearson correlation coefficient (PCC)**, which measures linear association between predicted and human scores across phoneme tokens, (ii) **Spearman correlation coefficient (SCC)**, which measures monotonic association and is reported alongside PCC when linearity is not assumed, (iii) **mean absolute error (MAE)**, and (iv) **mean squared error (MSE)**, which quantify magnitude mismatch after mapping predictions to $[0, 2]$. In confirmatory testing, raw PCC on canonical LPP tests H1; calibrated MAE/MSE supplement interpretation but do not define hypothesis rejection for H1.

Turning to calibration and leakage control, affine maps and learned scorers are fit using training speakers only. Predictions on the official test partition never influence parameter estimation, early stopping, or model selection. Clip operations enforce the human score range after calibration.

Because phoneme tokens within an utterance and speaker are dependent, point estimates alone can overstate precision. This thesis adopts (i) **percentile bootstrap confidence intervals** ($n_{\text{boot}} = 1{,}000$) for PCC, SCC, MAE, and MSE, (ii) **paired** ΔPCC on identical test phones when comparing two locked configurations, and (iii) **multi-seed checks** on selected learned models to ensure observed gaps exceed optimization noise.

Operational definitions and hypothesis-specific testing rules appear in Section 3.11 of Chapter 3. The present section establishes the metric vocabulary used when reading prior empirical studies (Section 2.9) and the results chapters.

2.9 Related Empirical Studies

This section surveys empirical work closest to the present factorial GOP study. The goal is not a complete bibliography but positioning: prior papers change multiple pipeline stages at once; this thesis isolates factors on Speechocean762.

Classical and Transition-Aware GOP. In early deep neural acoustic modeling, senone posteriors replaced GMM likelihoods, but HMM state transition probabilities (STPs) were frequently omitted because each senone is shared across many states. Sudhakara et al. [15] derived a DNN-HMM GOP formulation that multiplies senone posteriors with STPs and implemented the measure in Kaldi. On non-native English collected from Indian learners, the transition-aware scores correlated more strongly with expert ratings than posterior-only baselines, indicating that the omitted transition term is not always negligible.

Self-Supervised and Alignment-Free GOP. Cao et al. [2] introduce alignment-free GOP formulations for CTC-trained models, addressing span-boundary sensitivity that affects traditional forced-alignment GOP on non-native speech. Related SSL-based assessment work demonstrates that fine-tuned or adapted Wav2Vec 2.0 representations improve correlation with human ratings on learner corpora; those studies differ in training protocol from the off-the-shelf comparison enforced here [10].

Building on alignment-free CTC formulations, Parikh et al. [11] address the high

computational cost of unrestricted substitution graphs, which scale poorly with phoneme sequence length and inventory size, by introducing Restricted Phoneme Substitutions (RPS). RPS constrains candidate substitutions with phonological clusters and empirical L2 error maps, and the authors evaluate the restricted graphs on Speechocean762 and a child-speech corpus. To mitigate overconfidence and gradient saturation in softmax-normalized posteriors, Parikh et al. [12] compare raw logit-based metrics such as *GOPMaxLogit* and *GOPMargin*. Peak logit values aligned more closely with human phoneme ratings on Speechocean762 than probability-based GOP, although classification accuracy still depended on the corpus. Removing the requirement for frame-synchronous ASR models altogether, Wong and Chen [17] derive phoneme posteriors from confusion networks (CN-GOP) built from Whisper N-best hypotheses. Cross-attention then integrates per-phoneme features with per-frame self-supervised representations without phoneme time alignments, and the method is evaluated on Speechocean762 and low-resource Tamil.

Learned Multi-Aspect Assessment. GOPT [7] reports strong multi-granularity performance using Transformer layers over LPP+LPR profiles. Ryu et al. [13] jointly optimize pronunciation assessment and mispronunciation detection with multi-task learning on Speechocean762, showing correlation between proficiency scores and phonetic errors. Shekar et al. [14] combine Wav2Vec 2.0 mispronunciation detection with multi-level GOP Transformers for intelligibility assessment on ITA speech, adjacent in method but different in corpus and task framing.

Extending parallel modeling paradigms, Yan et al. [18] introduce *HierTFR*, a hierarchical Transformer that models the phone, word, and utterance levels together with inter-aspect correlations. Aspect attention, selective fusion, and a correlation-aware regularizer are trained on Speechocean762, together with level-specific pretraining strategies. The design improves multi-aspect scoring relative to parallel GOPT-style baselines, but it still reports a bundled system rather than a locked one-factor comparison.

Data Augmentation and SSL Benchmarks. Fort and Tyers [5] evaluate Wav2Vec2-BERT for computer-assisted pronunciation training in a low-resource language setting, illustrating how checkpoint and fine-tuning data affect mispronunciation detection. El Kheir et al. [8] pursue end-to-end joint recognition and diagnosis without explicit GOP features; such systems complement rather than duplicate the GOP factorization pursued here.

To address the scarcity of annotated L2 training speech, recent studies introduce targeted data augmentation. Fu et al. [6] propose a phone-level mixup that constructs artificial word-level feature representations by sampling phone-level vectors and GOP scores from an acoustic pool, combined with multi-source acoustic inputs (MFCC and deep features). On Speechocean762, mixup pretraining raised word-level correlation with a smaller unlabeled pool than the real-data baseline. Addressing signal-level continuity, El Kheir et al. [9] develop *SpeechBlender*, a fine-grained audio augmentation pipeline that uses blending masks and mixing factors to interpolate raw speech between candidate and donor phonemes. The method synthesizes realistic mispronunciation classes in low-resource settings and improved phoneme-level scoring on Speechocean762 and Arabic AraVoiceL2. Both augmentation lines enrich training data for learned scorers; they do not isolate GOP representation, alignment, or acoustic checkpoint under a protocol lock.

Taken together, these studies report valuable point estimates but rarely lock dataset partition, alignment artifact, GOP representation, acoustic checkpoint, and scorer while varying a single factor. The empirical gap is therefore not the absence of neural CAPT models but the absence of a *reproducible factorial benchmark* on a standard non-native corpus with expert phoneme scores. Section 2.10 synthesizes that literature-based gap explicitly; the methodology and results chapters execute the comparison under protocol lock on Speechocean762 [19].

2.10 Literature-Based Gap

Sections 2.1–2.9 reviewed CAPT evolution, GOP formulations, acoustic-model trends, datasets, metrics, and representative empirical papers. This section synthesizes the literature-based gap: three shortcomings in prior work that justify the controlled factorial study

undertaken in this thesis.

Representational ambiguity. The label "GOP" is applied to canonical LPP, competitive LPR, low-dimensional vectors, full confusion profiles, span-based scores, and alignment-free graph scores [16, 2, 7]. Without factorial comparison on identical phoneme tokens, it remains unclear which geometric aspect drives agreement with experts.

Confounded comparisons. Improved correlation after adopting Wav2Vec 2.0 [1, 3] or Transformer scorers [7] may stem from any combination of encoder, inventory, alignment, GOP formula, and learned head. Published benchmarks seldom hold the Speechocean762 partition and phoneme evaluation unit fixed while changing one factor, which limits causal interpretation [19].

Limited failure analysis. Global PCC can hide phoneme-class effects, speaker instability, and structured residuals tied to alignment or accent markup [19]. Few papers publish stratified tables and residual taxonomies on the same locked predictions used for confirmatory tests.

Thesis positioning. In response, this thesis conducts a protocol-locked factorial evaluation on Speechocean762. RQ1 establishes a Kaldi GOP baseline; RQ2 isolates representation effects; RQ3 compares acoustic-model posteriors; and RQ4 examines behavioral variation and failure modes across phonemes and speakers. Learned scorers extend the confirmatory design as a secondary modeling layer.

Chapter 3 operationalizes the protocol; Chapters 4 and 5 report whether the confirmatory hypotheses hold and how GOP behaves in practice.

Chapter 3
Research Methodology

3.1 Research Design

This study is a controlled factorial evaluation of GOP scoring. Independent factors are acoustic model, feature representation, decoding graph, and scoring function. Traditional GOP conflates those stages; the design separates them into a baseline, representation, acoustic-model, behavioral, learned-scoring, and verification phase.

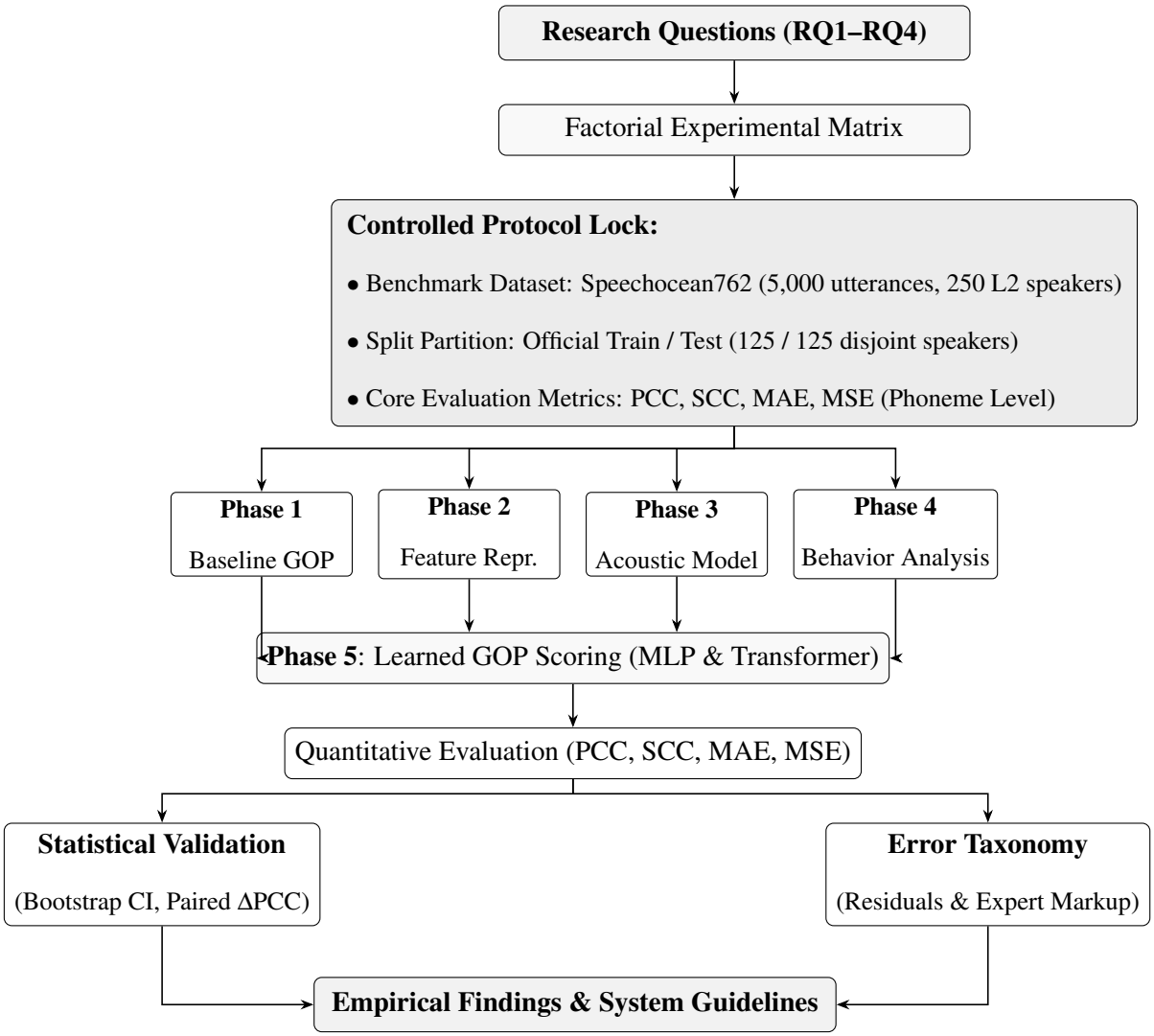


Figure 3.1: Overall Research Design and Methodological Workflow

The workflow in Figure 3.1 maps research questions onto a factorial matrix under a locked protocol. The investigation establishes a Kaldi canonical LPP baseline, then varies representation, acoustic-model posteriors, and scoring architecture, and finally stratifies behavior and residuals. Learned scorers operate on frozen features. Bootstrap intervals, paired ΔPCC, and residual taxonomy verify the confirmatory claims.

Table 3.1: Systematic Experimental Factors and Controlled Configurations

Factor	Investigated Levels	Controlled Baseline	Target Objective
Acoustic Model	Kaldi M13 (TDNN-F Chain), Wav2Vec2-XLSR-53 (espeak CTC), Wav2Vec2-lv60 (espeak CTC)	Kaldi M13 (TDNN-F Chain Baseline)	Quantify impact of acoustic representations on posterior quality
Feature Representation	Canonical LPP, Competitor LPR, Rank-2 Vector, 84-d / 78-d Confusion Profiles	Canonical LPP scalar	Identify optimal geometric representation for phonetic error detection
Alignment & Graph	Kaldi Forced Alignment, Alignment-Free Substitution (AF-S), Substitution-Deletion (AF-SD)	Fixed Kaldi CTM time boundaries	Disentangle boundary alignment errors from phonetic substitution and deletion
Scoring Architecture	Direct Mapping, Multivariate OLS, Feed-Forward MLP, Sequence Transformer	Univariate Train-Only Linear Mapping	Measure capacity of nonlinear and contextual modeling on score correlation
Phoneme Embedding	Canonical Dense Embedding (42-way Kaldi, 39-way SSL) vs. No Embedding	Without Phonetic Identity Embeddings	Measure performance gains from explicit phonetic canonical prior
Subgroup Stratum	Vowel vs. Consonant, Individual Speakers ($n = 125$), Sentence-Score Tertiles	Aggregate Test Split ($n = 47,369$)	Evaluate fairness, variance, and calibration across demographic and phonetic subsets

Every comparison holds the Speechocean762 test tokens, speaker-disjoint split, and metric script invariant (Table 3.4). Scoring models are fit on the training partition only. Observed

ΔPCC therefore reflects the isolated experimental factor. The operational matrix is Table 3.2.

Table 3.2: Comprehensive Experimental Matrix Across Investigation Phases

Investigation Phase	Experimental Configuration	Independent Variable	Primary Purpose
Baseline GOP Scoring	Kaldi Canonical LPP Extraction	Baseline Pipeline	Extract canonical GOP
	Direct Linear Scoring vs. Human	Evaluation on Test Split	Establish reference baseline
GOP Representation	Standard GOP vs. Canonical LPP	Feature Identity Check	Verify extract equivalence
	LPR vs. Best Acoustic Competitor	Relative Competition	Assess competing phone signal
	GOP-Only Rank-2 Vector (OLS)	Compact Multivariate Vector	Linear combination test
	84-dimensional LPP+LPR Matrix	Full Confusion Profile	Dense geometric modeling
Acoustic Model & Graph	Direct Kaldi Baseline Reference	Acoustic Model (Kaldi)	Control verification
	Wav2Vec2-XLSR-53 Cao GOP-S	Pre-trained Multilingual SSL	Alignment-free substitution
	Wav2Vec2-lv60 Cao GOP-S	Pre-trained Monolingual SSL	Encoder capacity comparison
	Wav2Vec2-XLSR-53 Cao GOP-SD	Substitution-Deletion Graph	Deletion modeling on XLSR
	Wav2Vec2-lv60 Cao GOP-SD	Substitution-Deletion Graph	Deletion modeling on lv60
Behavioral Analysis	Vowel vs. Consonant Stratification	Phonetic Category	Assess broad phonetic stability
	Speaker-Level Grouping ($n = 125$)	Speaker Identity	Measure inter-speaker spread
	Sentence Accuracy Tertiles	Performance Stratum	Evaluate ceiling/floor effects
Learned GOP Scoring	Kaldi compact GOP vector + MLP / Transformer	Scoring Architecture	Nonlinear & sequence gain
	SSL GOP-S / GOP-SD + MLP / Transformer	Learned Scoring on SSL GOP	Validate learned SSL GOP
	Kaldi / SSL Matrices + MLP / Trans.	Matrix Representation	Compare dense feature inputs
	Matrix + Canonical Phone Embedding	Phonetic Prior Embedding	Evaluate explicit identity prior
Statistical Validation	1,000-Fold Bootstrap CI Estimation	Non-parametric Resampling	Quantify metric uncertainty
	Paired bootstrap ΔPCC (same phones)	Hypothesis Contrast	Confirm distinguishable gaps
	Multi-Seed Stability (5 Seeds)	Training Random Seed	Assess optimization variance
	Residual & Expert Markup Taxonomy	Error Categorization	Diagnose failure modes

3.2 Hypothesis Operationalization

Section 3.1 established a controlled factorial design in which every comparison is executed under a locked protocol: the Speechocean762 corpus, the official speaker-disjoint train/test partition (125/125 speakers), phoneme-level evaluation, and the metric suite PCC, SCC, MAE, and MSE. The four research questions (RQ1-RQ4) and three confirmatory hypotheses (H1-H3) are stated in Section 1.5 (Chapter 1). This section explains why each question is necessary, specifies how H1-H3 are tested in the rule below, and maps questions to experimental phases in Table 3.3. The questions follow the three-tier logic of a pure empirical study: whether GOP reflects expert ratings; which factors change that validity; and where GOP fails when examined by phoneme and speaker.

RQ1-RQ3 are confirmatory: each is paired with a directional hypothesis that can be rejected using correlation and paired difference tests (Section 1.5), following the testing rule below. RQ4 is exploratory and carries no directional H4, because its aim is to describe assessment behavior and to taxonomize errors rather than to test a single parameter. Learned GOP scoring (MLP and Transformer) is not a separate research question; it is a modeling layer that extends RQ1 (the validity of mapping GOP to human scores) and RQ2 (GOP representation under a learned scorer). Statistical validation and residual taxonomy (bootstrap confidence intervals, paired ΔPCC, multi-seed checks) form a shared verification layer for all four questions, not an independent question.

Rationale for the Research Questions.

RQ1 is a locked baseline: classical GOP [16] must be measured on Speechocean762 before later gains can be attributed. H1 requires only a positive association, not a deployment-grade PCC.

RQ2 treats GOP as a family of geometries (LPP, LPR, rank-2 vector, GOPT-style profile; Section 2.6) rather than a unique scalar. The Kaldi extract is locked so that representation is the only confirmatory factor.

RQ3 treats GOP as a function of $P(p \mid x_t)$. Kaldi versus SSL-CTC is a sensitivity analysis

of the posterior source, not a matched bake-off [3, 1, 2]. A follow-up freezes each SSL encoder and changes only substitution versus substitution-deletion.

RQ4 is exploratory. Pooled PCC can hide phoneme, speaker, and score-stratum structure. The corpus has no CEFR labels, so tertiles of sentence accuracy are an empirical index, not a proficiency gold label [19].

These rationales are tested through a locked inferential rule.

Hypothesis Testing Strategy. Primary inference is a percentile bootstrap 95% CI ($n_{\text{boot}} = 1{,}000$) and paired ΔPCC on the same phones (Section 3.11). H1 is tested in raw GOP space; H2 and H3 in calibrated $[0, 2]$. A difference is not claimed if the CI of Δ contains zero.

H1 rejects $\rho = 0$ if the two-sided CI of raw Kaldi LPP PCC lies entirely above zero. H2 is supported if at least one of the six linear representation pairs excludes zero; learned scorers do not enter H2. H3 uses the three direct-scoring pairs in Table 3.12. Multi-seed checks apply only to two representative learned models and are not tests of H1-H3.

Mapping Research Questions to Experiments. Table 3.3 maps each research question to its hypothesis, to the investigation phase of Figure 3.1 and Table 3.2, and to the analysis used to answer it.

Table 3.3: Mapping of research questions to hypotheses, experimental phases, and analyses

Question	Hypothesis	Experimental phase	Independent factor	Primary analysis
RQ1 GOP–human validity	**H1** $\rho > 0$	Baseline GOP: canonical Kaldi LPP versus expert scores on the official test	Traditional GOP pipeline (all else locked)	PCC, SCC on raw GOP; MAE, MSE after train-only map; bootstrap CI of PCC
RQ1 (modeling extension)	- (no new H)	Learned GOP scoring: MLP / Transformer on frozen features	Scoring architecture (linear / MLP / Transformer; $\pm$ phone embedding)	PCC on predictions clipped to $[0,2]$; comparison with direct / OLS on the same features; multi-seed on two representative models
RQ2 Representation	**H2** PCC differs by representation	GOP representation: LPP, LPR, rank-2 vector, 84-d profile	GOP feature	Per-representation PCC/SCC; paired ΔPCC
RQ2 (extension)	-	Learned scoring on LPP+LPR profiles (Kaldi 84-d / SSL 78-d) $\pm$ phone embedding	Representation $\times$ scorer	Interpreted only after linear H2; no PCC comparison across unlike feature types
RQ3 Acoustic model	**H3** PCC differs by AM	Acoustic model: Kaldi M13 (TDNN-F Chain); Wav2Vec2-XLSR-53; Wav2Vec2-lv60 (alignment-free substitution GOP, no fine-tuning)	Posterior source (acoustic model)	Paired ΔPCC: XLSR-53$--$Kaldi, lv60$--$Kaldi, XLSR-53$--$lv60; bootstrap CI
RQ3 (graph follow-up)	-	Same XLSR-53 or lv60 encoder; substitution $\rightarrow$ substitution–deletion	Decoding graph	ΔPCC within each encoder-locked pair
RQ4 Behavior	*No H4* (exploratory)	Behavioral analysis: phoneme, speaker, sentence–score tertiles	Stratum (phone / speaker / score band)	Stratified PCC/SCC/MAE; minimum-N threshold; no CEFR inference
RQ4 (failure modes)	*No H4*	Error analysis on frozen residuals	- (taxonomy on locked predictions)	T1–T5 residual taxonomy (Section 4.8); expert markup
RQ1-RQ3 (verification)	H1–H3	Statistical validation	- (models unchanged)	Bootstrap CI; paired ΔPCC; multi-seed

Each confirmatory row is permitted to vary only one independent factor, except the H3 contrasts against Kaldi, which change both the posterior source and the GOP formula and are interpreted as sensitivity analyses. Extension and follow-up rows answer the same question after the protocol lock has been relaxed; they are not pooled into a single "better than" claim with the confirmatory row. Learned scoring is placed under RQ1 because the originating question is the validity of mapping GOP to human scores, not the claim that "the MLP improved GOP itself".

3.3 Dataset

GOP-based pronunciation assessment is only as interpretable as the human scores against which it is compared. A sentence-level rating cannot locate a substitution, a deletion, or an accented but acceptable realization, which are precisely the events that phoneme-level GOP is designed to quantify. This study therefore uses a single evaluation corpus that supplies expert phoneme scores, a documented L2 speaker population, and an official speaker-disjoint partition: Speechocean762 [19]. All GOP-human agreement figures reported in later chapters are computed on this corpus. The two Wav2Vec 2.0 checkpoints compared in the acoustic-model phase (XLSR-53 and lv60) are used off-the-shelf and are not trained or fine-tuned on Speechocean762.

Speechocean762 comprises 5,000 English utterances from 250 non-native speakers (20 utterances per speaker), recorded at 16 kHz [19]. Five expert raters provide annotations at phoneme, word, and sentence granularity. The primary dependent variable in this thesis is the phoneme score; word scores are not used as a headline metric. Sentence-level accuracy is retained only to construct the speaker tertiles of RQ4 (Section 3.2), not as a proficiency label.

Table 3.4: Locked Speechocean762 evaluation set used in every experimental phase

Split	**Speakers**	**Utterances**	**Phoneme tokens**
Train	125	2,500	47,076
Test	125	2,500	47,369
Total	250	5,000	94,445

Table 3.4 records the partition that implements the protocol lock of Section 3.1. The official train and test sets each contain 125 speakers and 2,500 utterances. After silence and noise symbols are removed, the locked extract yields 47,076 scored phoneme tokens on train and 47,369 on test. Speaker overlap between the two splits is zero: no speaker contributes utterances to both fitting and evaluation. This study does not redraw the split, does not shuffle utterances across speakers, and does not report headline PCC, SCC, MAE, or MSE on any partition other than the official test set. When a learned scorer requires a validation set, speakers are carved from the *training* side only, under a speaker-independent rule with a fixed seed; the official test set is never used for fitting, early stopping, or model selection.

Table 3.5: Human score scales used in this study

Annotation	**Range**	**Role in this thesis**
Phoneme score	0–2	Primary dependent variable (RQ1-RQ3). 2 = correct; 1 = correct but heavily accented; 0 = wrong or missing. Scores are rounded with a floor of 0.1.
Sentence accuracy	0–10	Speaker-mean tertiles for the behavioral analysis of RQ4 only. Not treated as a CEFR or proficiency gold label.

Table 3.5 states how each available scale is used. Speechocean762 does *not* provide CEFR bands or Beginner / Intermediate / Advanced labels. Speaker age metadata, where present, is descriptive and is not used as a proficiency proxy. Consistent with the decision that RQ4 carries no H4, this study does not infer proficiency from scores and then treat that inference as ground truth.

The unit of analysis is the phoneme token, identified by utterance identifier and intra-utterance phone index. Canonical phone symbols follow a stress-stripped 39-phone CMU inventory (for example `IY0` is mapped to `IY`). Silence and noise symbols (`SIL`, `SPN`, `NSN`) are excluded. On the locked extract, every retained token has a human phoneme score (zero missing labels) and no silence token remains after filtering. The 42-slot Kaldi LPP layout and the CMU-to-espeak IPA mapping used by the SSL checkpoints are specified with preprocessing and acoustic models, not in this section.

Detailed expert markup (accent versus substitution or confusion) is available alongside the numeric scores. That markup is not a training target. It is reserved for the qualitative residual taxonomy that accompanies statistical validation. Multi-task word- or utterance-level scoring heads are outside the scope of the study: GOP-human agreement is reported only at the phoneme level on Speechocean762.

With the corpus, split, evaluation unit, and score scales locked, the next section specifies how the audio is preprocessed before alignment and posterior extraction.

3.4 Data Preprocessing

Section 3.3 locked *which* tokens are evaluated. This section locks *how* raw Speechocean762 recordings and annotations are turned into those tokens. Preprocessing is not an experimental factor: every investigation phase reads the same 16 kHz waveform, the same canonical phone symbol, the same human score, and the same join key. Subsequent sections vary only the acoustic model, the GOP representation, or the scoring function. No speed perturbation, additive noise, or SpecAugment is applied; no additional speaker- or utterance-level filters are imposed beyond silence removal.

Locked preprocessing therefore consists of four shared steps: (i) a mono floating-point waveform at 16 kHz, (ii) a join key of utterance identifier plus intra-utterance phone index (for example `000010011.0`), (iii) canonical phones stress-stripped onto the 39-phone CMU inventory (`IY0` becomes `IY`), and (iv) a silence and noise filter that drops `SIL`, `SPN`, `NSN`, or `<eps>` (case-insensitive). Multi-channel files, if encountered, are reduced by averaging channels. Files whose native rate differs from 16 kHz are resampled to that rate. Speechocean762 is already distributed at 16 kHz, so resampling is a protocol constraint rather than a corpus conversion [19]. Convolutional hop sizes and SSL feature-extractor settings belong with the acoustic models, not here. The same stress-stripping rule is applied later to expert markup so that residual taxonomy uses the identical symbol set. After the silence filter the locked extract contains 47,076 training tokens and 47,369 test tokens, with zero remaining silence symbols and zero missing human phoneme scores (Table 3.4). Human phoneme scores occupy the 0-2 scale of Table 3.5. Numeric rounding uses a floor

of 0.1. When a scoring function maps GOP into the human range, predictions are clipped to $[0,2]$; the mapping itself is specified with pronunciation scoring and evaluation metrics.

Table 3.6 summarizes the rules that every phase inherits.

Table 3.6: Locked preprocessing steps applied before GOP computation

Step	**Rule**	**Applies to**
Waveform	16 kHz, mono, floating point	Kaldi extract and SSL inference
Join key	Utterance identifier + phone index	All phases
Phone symbol	39 CMU phones, stress stripped	All phases
Silence filter	Drop `SIL`, `SPN`, `NSN`, `<eps>`	All phases
Human score	Rounding floor 0.1; clip to $[0,2]$ after mapping	Evaluation
Kaldi layout	42 LPP slots and 42 LPR slots; 0-based `phone_id`	Baseline and representation phases
SSL phone identity	One-to-one CMU $\rightarrow$ espeak IPA map (39 tokens; no silent fallback)	Wav2Vec2-XLSR-53 and Wav2Vec2-lv60
CTM time stamps	Scale existing start/end times to waveform duration	SSL frame indexing; not a new aligner

Turning to the acoustic-model-specific extracts, the Kaldi and SSL branches inherit the same tokens but not the same feature layout.

Kaldi Feature Layout. The Kaldi M13 branch reads the precomputed 85-dimensional feature extract publicly released with the GOPT repository (`https://github.com/YuanGongND/gopt`) [7], which was generated by the official Kaldi GOP recipe for Speechocean762 [19]:

$$\big[\texttt{phone_id},\ \mathrm{LPP}_0,\ldots,\mathrm{LPP}_{41},\ \mathrm{LPR}_0,\ldots,\mathrm{LPR}_{41}\big].$$

The 42 posterior slots comprise three un-scored symbols (`SIL`, `SPN`, `NSN`) and the 39 CMU phones. The stored `phone_id` is already a 0-based index into the LPP block (typically 3 through 41); an off-by-one index collapses the GOP-human correlation and is treated as

a pipeline failure. Within the Kaldi recipe, frame-level posteriors of tied states (senones) belonging to the same pure phone are summed at each frame, and LPP is computed as the mean log phone posterior across the aligned phone span. LPR columns are parsed so that the column layout is locked; whether LPR outperforms LPP is evaluated in the GOP-representation section using this same extract.

CMU-to-espeak Identity. Wav2Vec2-XLSR-53 and Wav2Vec2-lv60 emit espeak IPA tokens rather than CMU symbols [3, 1]. Before GOP is computed, each of the 39 scored CMU phones is mapped to exactly one espeak token. The map is one-to-one: a missing CTC vocabulary entry is a hard error, not a silent substitution. Alignment-free GOP keeps the full espeak-plus-blank softmax ($V = 392$); that likelihood is defined with GOP computation. This section only guarantees that the canonical phone already has a unique CTC identity when scoring begins.

CTM-to-Waveform Matching. Many Speechocean762 waveforms are shorter than the accompanying phone CTM time stamps (median duration ratio approximately 0.385). Before SSL frames are indexed, every CTM start and end is multiplied by $\text{duration}_{\text{wav}}/\text{duration}_{\text{CTM}}$ so that frame indices fall on the 16 kHz waveform. This is a duration scale, not a new forced-alignment algorithm. How the CTM itself is produced is specified in Section 3.5.

No encoder is trained or fine-tuned on Speechocean762 as part of preprocessing, and the phone inventory is not changed between experimental phases. With waveforms, symbols, join keys, and extract layouts locked, Section 3.5 states the origin of the time boundaries themselves.

3.5 Forced Alignment

Phoneme-level GOP requires a canonical phone identity in utterance order. The same alignment artifact can supply that identity in two ways: as a time span (t_s, t_e) over which posteriors are pooled, or as a silence-stripped phone sequence y on which alignment-free

GOP is computed without using Kaldi start and end times as the scoring window [2]. This section locks the artifact. Whether a formula uses spans or only y is specified with GOP computation.

Locked Kaldi CTM. Alignment is taken from the phone-level CTM released with the Kaldi M13 GOP extract. Each line records utterance identifier, channel, start time, duration, and a possibly stressed phone (for example `IY0`). After silence is removed, token order matches the Speechocean762 phoneme annotations under the join key of Sections 3.3 and 3.4.

The Kaldi baseline reads precomputed LPP and therefore does not re-open start/end times at evaluation. The CTM remains the sole source of the canonical sequence y for the SSL branch. One CTM is used in every experimental phase. Changing the acoustic model (Kaldi M13, Wav2Vec2-XLSR-53, or Wav2Vec2-lv60) does not change the CTM; switching an alignment-free substitution graph to a substitution-deletion graph does not change it either.

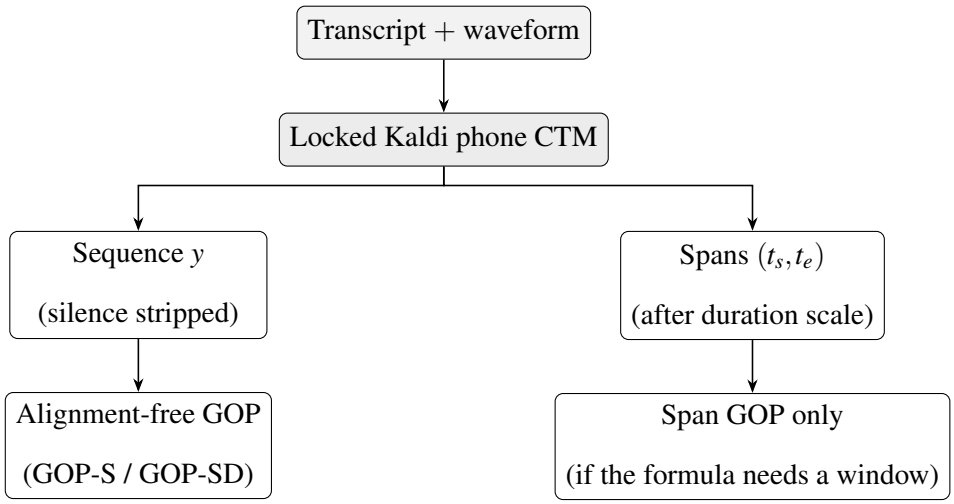


Figure 3.2: Kaldi CTM as the single alignment source: phone sequence y for alignment-free GOP, and time spans only when a formula requires a Kaldi window.

Figure 3.2 states the split. Table 3.7 records what each object is allowed to mean.

Table 3.7: Alignment artifacts and their role under the protocol lock

Artifact	**Content**	**Used as**	**Not used as**
Kaldi phone CTM	Start, duration, stressed phone	Sole source of y and, when needed, of time spans	An aligner re-run in this thesis
Sequence y	CTM after dropping silence and stress	Input to GOP-S / GOP-SD; CTC Viterbi for SSL 78-d profiles	A human score
Spans (t_s, t_e)	Start/end after the duration scale of Section 3.4	Only if the GOP formula needs a Kaldi window	The scoring window for alignment-free XLSR-53 / lv60 GOP

The CTM is cleaned before either branch is used: (i) Kaldi position markers (`_B`, `_E`, `_I`, `_S`) are stripped when present, (ii) stress is removed exactly as in Section 3.4 (`IY0` $\rightarrow$ `IY`), and (iii) silence and noise symbols (`SIL`, `SPN`, `NSN`, `<eps>`) are dropped. The resulting sequence length equals the scored-token counts already reported (47,076 train, 47,369 test).

Two Time Branches. To use this locked CTM without introducing a second aligner, the pipeline splits into a Kaldi span branch and an alignment-free sequence branch. On the Kaldi baseline and representation phases, LPP is already aggregated inside the 85-dimensional extract. This section does not reconstruct `ali-to-phones`; senone-to-phone pooling is an attribute of the Kaldi M13 recipe and is treated with the acoustic model and GOP formula.

On Wav2Vec2-XLSR-53 and Wav2Vec2-lv60, headline GOP is alignment-free over y [2]. Kaldi start and end times are not the scoring window. Learned 78-dimensional LPP+LPR profiles on those encoders pool posteriors along a CTC Viterbi path of the *same* y, not along Kaldi spans; that pooling is specified with GOP representations and learned scoring. The CTM-to-waveform duration scale (median ratio approximately 0.385) is already defined in Section 3.4: it matches clocks, and is not a second aligner.

With the alignment source locked, Section 3.6 specifies the three acoustic models that emit the posteriors from which GOP is computed.

3.6 Acoustic Models

GOP is a function of the phoneme posterior $P(p \mid x_t)$, or of a pre-aggregated log phone posterior. Changing the acoustic model therefore changes GOP even when the scoring formula is written identically. The three models below answer RQ3/H3: whether GOP-human agreement depends on the posterior source. The comparison measures that sensitivity; it is not a matched-architecture ranking and does not assert *a priori* that SSL outperforms Kaldi.

When the acoustic model varies, the protocol of Sections 3.3–3.5 is held fixed: Speechocean762, the official split ($n_{\text{test}} = 47{,}369$), 16 kHz audio, the 39 scored CMU phones, the locked Kaldi CTM, and direct scoring with a train-only map. The independent factor for H3 is the acoustic model. The alignment-free substitution versus substitution-deletion graphs freeze each SSL encoder and change only the decoding graph; that follow-up belongs with GOP computation, not as a fourth model.

Table 3.8: Acoustic models used to generate phoneme posteriors

Model	**Type**	**Posterior**	**Phone interface**
Kaldi M13 (TDNN-F Chain)	Factored TDNN, chain objective	Canonical LPP, 42 slots (39 scored)	CMU (senone $\rightarrow$ phone in the extract recipe)
Wav2Vec2-XLSR-53 espeak CTC	Multilingual SSL + CTC	Frame softmax over espeak + blank ($V = 392$)	One-to-one CMU $\rightarrow$ IPA (Section 3.4)
Wav2Vec2-lv60 espeak CTC	English wav2vec 2.0 Large + CTC	Same espeak CTC interface	Same map as XLSR-53

Table 3.8 lists the checkpoints `facebook/wav2vec2-xlsr-53-espeak-cv-ft` and `facebook/wav2vec2-lv-60-espeak-cv-ft` [3, 1].

The Kaldi M13 baseline is the pretrained Librispeech ASR Chain 1d model (`https://kaldi-asr.org/models/m13`, Kaldi version `ea6e1b7`, released in 2020), a factored time-delay neural network (TDNN-F) trained with the chain objective on 960 hours of Librispeech with 3-fold speed perturbation (recipe `tdnn_1d_sp`) using 40-dimensional high-resolution

MFCCs and 100-dimensional online i-vectors. Its native outputs are senone (tied-state) distributions. Senone-to-phone pooling is performed in the Kaldi M13 extract recipe; this thesis reads the resulting 42-dimensional LPP of Section 3.4 and does not re-run Kaldi on Speechocean762. The same extract is the input to the GOP-representation phase.

The thesis uses this model as the baseline for three reasons. It is the traditional HMM-based posterior source against which the SSL checkpoints are contrasted in RQ3. It is released pretrained, so the posterior source stays fixed across all phases and no model is trained on Speechocean762. The Kaldi GOP recipe for Speechocean762 and the GOPT benchmark use the same model [19, 7], which allows the comparison with published results in Section 5.7. No second Kaldi acoustic model, such as a GMM-HMM or a separately trained TDNN, was evaluated, so conclusions about Kaldi GOP refer to this chain model only.

Both SSL systems share the Wav2Vec 2.0 encoder: a convolutional feature encoder maps the waveform $\mathscr{X}$ to latent units $\mathscr{Z}$, and a Transformer context network produces representations $\mathscr{C}$ [1]. Weights are held frozen at inference. Phoneme posteriors come from the espeak CTC head; $\mathscr{C}$ itself is not GOP. Each checkpoint is consumed with its released feature extractor at 16 kHz.

XLSR-53 and lv60 keep the same CTC head family, the same CMU-to-IPA map, the same alignment-free substitution GOP, and the same canonical sequence y from Section 3.5. They differ in encoder training data and capacity (multilingual XLSR versus English wav2vec 2.0 Large). The planned H3 contrasts are therefore XLSR-53-Kaldi, lv60-Kaldi, and XLSR-53-lv60. The CTC vocabulary is not changed between the two SSL checkpoints.

On the alignment-free branch the frame softmax retains the full espeak inventory and the blank [2]. It is not renormalized onto the 39 CMU phones. The canonical identity is the mapped espeak token of Section 3.4. GOP-S and GOP-SD formulae, and the graph follow-up that holds each encoder fixed, are specified next.

3.7 GOP Computation

This section states the three scalar GOP formulae used in the study. The extract layout, phone map, CTM sequence y, and acoustic models are already specified in Sections 3.4–3.6. The same GOP-S and GOP-SD formulae are applied to Wav2Vec2-XLSR-53 and Wav2Vec2-lv60.

Classical GOP averages the log posterior of the canonical phone over the aligned span [16]. That frame mean is not recomputed from the waveform. The Kaldi M13 extract already stores a 42-slot log phone posterior; the score is the canonical entry

$$\mathrm{GOP}(p) = \mathrm{LPP}[p], \tag{3.1}$$

where p is the 0-based `phone_id` into the LPP block of Section 3.4.

On the SSL branch, GOP is the alignment-free CTC log-likelihood ratio of Cao et al. [2]. The input is the silence-stripped canonical sequence y (espeak identities from Section 3.4), not the Kaldi span (t_s, t_e):

$$\text{GOP-S}(y_i) = \log P(y \mid x) - \log P\big(y_{\mathrm{S}}^{(i)} \mid x\big). \tag{3.2}$$

The sequence $y_{\mathrm{S}}^{(i)}$ replaces position i by a substitution wildcard: y_i is not forced. The frame softmax retains the blank and the full espeak vocabulary ($V = 392$) and is not renormalized onto the 39 CMU phones.

GOP-SD uses the same numerator as GOP-S. The denominator additionally allows deletion of y_i:

$$\text{GOP-SD}(y_i) = \log P(y \mid x) - \log P\big(y_{\mathrm{SD}}^{(i)} \mid x\big), \tag{3.3}$$

where $y_{\mathrm{SD}}^{(i)}$ is a substitution-or-deletion wildcard (position i may be skipped).

3.8 GOP Representations

This section defines four representations on the same Kaldi 85-dimensional extract. The vector layout is given in Section 3.4; the scalar GOP formulae are given in Section 3.7.

On this extract, standard GOP coincides with the canonical log phone posterior:

$$\mathrm{GOP}(p) = \mathrm{LPP}[p]. \tag{3.4}$$

The stored LPR slots satisfy $\mathrm{LPR}[q] = \mathrm{LPP}[p] - \mathrm{LPP}[q]$, so the canonical entry $\mathrm{LPR}[p]$ is identically zero and is not used as a score. The competitor LPR is

$$\mathrm{LPR}(p) = \mathrm{LPP}[p] - \max_{q \neq p} \mathrm{LPP}[q]. \tag{3.5}$$

A two-component vector keeps the canonical posterior and the strongest competitor:

$$\left[\mathrm{LPP}[p],\ \max_{q \neq p} \mathrm{LPP}[q]\right]. \tag{3.6}$$

$\mathrm{LPR}(p)$ is exactly the difference of those two entries and is not appended as a third component.

The 84-dimensional profile concatenates the full LPP and LPR blocks,

$$\mathbf{v}_{84} = \left[\mathrm{LPP}_0, \ldots, \mathrm{LPP}_{41},\ \mathrm{LPR}_0, \ldots, \mathrm{LPR}_{41}\right]. \tag{3.7}$$

This is the GOPT-style profile (Section 2.6).

3.9 Pronunciation Scoring

This section details the methodologies used to transform extracted Goodness of Pronunciation (GOP) representations into human-calibrated pronunciation scores on the target scale of $[0, 2]$. The scoring framework encompasses three progressive paradigms: direct linear and multivariate ordinary least squares (OLS) projections, nonlinear

phoneme-independent Multi-Layer Perceptrons (MLP), and contextual sequence-level Transformer encoders. In addition, an explicit canonical phoneme embedding mechanism and a strict anti-leakage training protocol are defined.

Direct and Linear Scoring. Direct scoring provides the primary baseline by establishing an optimal linear mapping from raw acoustic GOP evidence to expert human scores without introducing complex learnable parameters.

For scalar GOP features (including canonical Kaldi LPP as well as Wav2Vec2-XLSR-53 and Wav2Vec2-lv60 GOP-S and GOP-SD), univariate linear regression is defined as:

$$\hat{y} = \text{clip}(a \cdot \text{GOP} + b, 0, 2), \tag{3.8}$$

where scalar parameters a and b are fitted via least squares exclusively on the training partition. The fitted mapping is subsequently applied invariantly to test tokens to prevent data leakage.

For multi-dimensional representations, multivariate ordinary least squares (OLS) regression is employed:

$$\hat{y} = \text{clip}(\mathbf{w}^T \mathbf{x} + b, 0, 2), \tag{3.9}$$

where $\mathbf{w} \in \mathbb{R}^{d_{\text{in}}}$ and $b \in \mathbb{R}$ are estimated on training data. This linear formulation is applied to three structured inputs:

(i) The two-component vector $\mathbf{x} = \left[\text{LPP}[p],\ \max_{q \neq p} \text{LPP}[q]\right]^T \in \mathbb{R}^2$;

(ii) The 84-dimensional Kaldi confusion profile $\mathbf{v}_{84} = [\text{LPP}_0 \ldots \text{LPP}_{41},\ \text{LPR}_0 \ldots \text{LPR}_{41}]^T \in \mathbb{R}^{84}$;

(iii) The 78-dimensional SSL confusion profile $\mathbf{v}_{78} = [\text{LPP}_0 \ldots \text{LPP}_{38},\ \text{LPR}_0 \ldots \text{LPR}_{38}]^T \in \mathbb{R}^{78}$, constructed from the 39 mapped IPA phones pooled over CTC Viterbi frame alignments.

Multi-Layer Perceptron Scoring. To capture nonlinear profile-to-score maps that OLS cannot represent, a Multi-Layer Perceptron (MLP) architecture is introduced. The MLP operates in a phoneme-independent manner, evaluating each phonetic segment purely from its local acoustic feature vector without accessing adjacent phonetic context.

The network architecture consists of an input projection, two fully connected hidden layers, and a scalar linear output layer:

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \tag{3.10}$$

$$\mathbf{h}_2 = \mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2), \tag{3.11}$$

$$\hat{y}_{\mathrm{raw}} = \mathbf{W}_3\mathbf{h}_2 + b_3, \tag{3.12}$$

where $\mathbf{W}_1 \in \mathbb{R}^{32\times d_{\mathrm{in}}}$, $\mathbf{W}_2 \in \mathbb{R}^{32\times 32}$, $\mathbf{W}_3 \in \mathbb{R}^{1\times 32}$, and $\mathbf{b}_1, \mathbf{b}_2, b_3$ denote bias parameters. The final predicted score is obtained by bounding the activation: $\hat{y} = \mathrm{clip}(\hat{y}_{\mathrm{raw}}, 0, 2)$.

Sequence-Level Transformer Encoder Scoring. Pronunciation evaluation by human experts is inherently contextual, influenced by neighboring articulatory transitions, co-articulation patterns, and speaker-level speaking rate. To incorporate broad utterance-level context into phoneme scoring, a sequence-level Transformer encoder is deployed [7].

Utterance sequences are structured as follows:

(i) **Sequence Padding and Masking:** An utterance containing T phonemes is zero-padded to a uniform maximum sequence length $T_{\max} = 50$, accompanied by a binary attention mask $m_t \in \{0, 1\}$ identifying genuine phoneme tokens ($m_t = 1$ for $t \leq T$, and $m_t = 0$ for padded positions).

(ii) **Input Projection and Positional Encoding:** Feature vectors $\mathbf{x}_t \in \mathbb{R}^{d_{\mathrm{in}}}$ are linearly mapped to model dimension $d_{\mathrm{model}} = 32$ and augmented with standard sinusoidal positional encodings $\mathbf{p}_t \in \mathbb{R}^{d_{\mathrm{model}}}$:

$$\mathbf{e}_t = \mathbf{W}_{\mathrm{in}}\mathbf{x}_t + \mathbf{b}_{\mathrm{in}} + \mathbf{p}_t. \tag{3.13}$$

(iii) **Encoder Architecture:** The encoded representations are processed by a 2-layer Transformer encoder. Each layer comprises 4 self-attention heads, a feed-forward inner dimension $d_{\text{ff}} = 64$, layer normalization, and a dropout rate of 0.10.

(iv) **Per-Phone Regression Head:** A shared linear projection layer maps the contextual representation $\mathbf{z}_t \in \mathbb{R}^{32}$ at each sequence position directly to a scalar score: $\hat{y}_t = \text{clip}(\mathbf{w}_{\text{out}}^T \mathbf{z}_t + b_{\text{out}}, 0, 2)$.

In contrast to multi-task frameworks, no auxiliary classification tokens (e.g., `[CLS]`) or word/utterance hierarchical prediction heads are attached, isolating the impact of sequence context on phoneme scoring.

Canonical Phoneme Identity Embeddings. Beyond sequence context, canonical phone identity supplies a phoneme-specific prior. To evaluate this prior, a learnable dense embedding lookup $\mathbf{E}(p) \in \mathbb{R}^{d_{\text{model}}}$ is incorporated into the scoring architectures:

$$\mathbf{E}(p) = \mathbf{W}_{\text{emb}} \mathbf{u}_p, \tag{3.14}$$

where $\mathbf{u}_p$ is a one-hot identity vector corresponding to:

(a) The 42-class Kaldi phone inventory for Kaldi-based models;

(b) The 39-class mapped IPA inventory (`ssl_index`) for Wav2Vec2-based models.

In the MLP configuration, $\mathbf{E}(p)$ is concatenated with the acoustic representation. In the Transformer encoder, $\mathbf{E}(p)$ is added directly to the projected token embedding in Equation (3.13) prior to self-attention.

Training Protocol and Optimization. To ensure fair comparisons across all scoring models and prevent optimistic evaluation bias, a rigorous training protocol is established:

(i) **Speaker-Independent Validation Split:** The official test set (125 speakers, 47,369 tokens) remains strictly sequestered. A validation set comprising 20% of training

speakers (25 out of 125 speakers, fixed under seed 0) is held out exclusively for model selection and early stopping.

(ii) **Feature Normalization:** Feature-wise Z-score normalization (zero mean, unit variance) is fitted solely on phoneme tokens of the training-fit speakers (the 100 speakers remaining after the validation hold-out) and applied invariantly to the validation and test partitions. Validation and test tokens never enter the scaler.

(iii) **Loss Function and Optimization:** Neural scoring models are trained by minimizing masked Mean Squared Error (MSE) over genuine phoneme tokens:

$$\mathscr{L} = \frac{1}{\sum_{t=1}^{T} m_t} \sum_{t=1}^{T} m_t \left(\hat{y}_t - y_t\right)^2, \tag{3.15}$$

where $y_t \in [0,2]$ denotes the expert human ground truth. Locked hyperparameters are given in Table 3.9.

Table 3.9: Locked learned-scoring hyperparameters

Item	Value
MLP	Two hidden layers of width 32 with ReLU; batch size 256 phoneme tokens
Transformer	$d_{\text{model}} = 32$, 4 heads, 2 layers, $d_{\text{ff}} = 64$, dropout 0.1; $T_{\max} = 50$; batch size 32 utterances
Adam	Learning rate 10^{-3}; weight decay 10^{-5}
Schedule	At most 40 epochs; early stopping patience 8 on validation MSE
Phone embedding	42-way Kaldi inventory; 39-way SSL mapped IPA
Output	$\hat{y} \in [0,2]$

3.10 Evaluation Metrics

Metrics are computed at the phoneme level on the official test partition. Two spaces are used: raw GOP (PCC and SCC for H1) and calibrated $\hat{y} \in [0,2]$ (MAE, MSE, and confirmatory PCC for H2, H3, and learned scoring).

Pearson correlation measures linear association:

$$r = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{N}(y_i - \bar{y})^2}}, \tag{3.16}$$

Spearman correlation uses the same form on fractional ranks $R(x_i)$ and $R(y_i)$. MAE and MSE after a train-only map are

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|, \qquad \mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2. \tag{3.17}$$

Table 3.10: Summary of quantitative evaluation metrics

Metric	Symbol	Domain	Optimal Target	Primary Analytical Role
Pearson Correlation	PCC/r	$[-1,1]$	$+1.0$	Linear score association
Spearman Correlation	SCC/ρ	$[-1,1]$	$+1.0$	Monotonic rank preservation
Mean Absolute Error	MAE	$[0,2]$	0.0	Average absolute prediction error
Mean Squared Error	MSE	$[0,4]$	0.0	Penalty on large residuals

Maps, scalers, and network weights are fit on training speakers only. Hypothesis-specific rejection rules are in Section 3.11.

3.11 Statistical Analysis

This section locks the inferential procedure. Inputs are locked GOP values and predictions on the official test set. Numerical metrics appear in Chapter 4. Residual counts appear there as well; only taxonomy rules are specified here.

Score Spaces and Bootstrap. The resampling unit is the phoneme token. Headline comparisons never mix raw GOP with calibrated $\hat{y}$, because clipping after a train-only map can shift PCC.

Table 3.11: Locked score spaces for hypothesis tests and model contrasts

Space	**Variable** x	**Role**
Raw GOP	Scalar GOP, LPP, or LPR before mapping	H1 (PCC/SCC and CI); identity check that standard GOP equals canonical LPP
Calibrated $[0,2]$	$\hat{y}$ after a train-only map, OLS, or learned scorer, then clipped	H2, H3, learned contrasts, MAE/MSE, headline model CIs

Each metric $m \in \{\text{PCC, SCC, MAE, MSE}\}$ receives a percentile 95% confidence interval from $n_{\text{boot}} = 1{,}000$ draws with replacement. The reported point value is m on the original sample. H1 is one-sided (PCC > 0), yet the interval is two-sided at 95%. H0 is rejected only if that interval lies entirely above zero.

Paired ΔPCC and Tests of H1-H3. Building on these locked score spaces, paired ΔPCC tests H1-H3 on identical join keys. Each bootstrap replicate reuses one index vector:

$$\Delta^{(b)} = m\big(x^{A}_{i^{(b)}}, y_{i^{(b)}}\big) - m\big(x^{B}_{i^{(b)}}, y_{i^{(b)}}\big). \tag{3.18}$$

A difference is not claimed if the 95% CI of Δ contains zero. No permutation p-value is reported. Bonferroni correction is not applied across the full model grid.

H1. Null: raw Kaldi LPP PCC is zero on the official test set. Reject if the two-sided 95% CI lies entirely above zero.

H2. Four Kaldi representations are mapped to $\hat{y} \in [0,2]$ using training data only: scalar LPP, competitor LPR, rank-2 OLS, and 84-dimensional OLS. The six planned pairs are LPR-LPP, rank-2-LPP, 84-d-LPP, rank-2-LPR, 84-d-LPR, and 84-d-rank-2. H2 is supported if at least one interval excludes zero. Learned scorers do not enter H2.

H3. Direct scoring after a train-only map. Planned pairs (Table 3.12) compare XLSR-53 GOP-S, lv60 GOP-S, and Kaldi LPP. Contrasts against Kaldi change both posterior source and formula. GOP-SD versus GOP-S, with the encoder frozen, is a follow-up and is not pooled into H3.

Table 3.12: Planned H3 paired contrasts (direct scoring, calibrated $[0,2]$)

Contrast	**Factor that can be attributed**
XLSR-53 GOP-S − Kaldi LPP	Posterior source, with the GOP formula that accompanies each model
lv60 GOP-S − Kaldi LPP	Same as above
XLSR-53 GOP-S − lv60 GOP-S	SSL encoder, with CTC espeak and GOP-S held fixed

Extension Contrasts and Residual Taxonomy. Consequently, learned contrasts, multi-seed checks, and the T1-T5 taxonomy remain outside the confirmatory tests of H1-H3. Learned contrasts use calibrated scores and the same feature X when a scorer or embedding effect is claimed. Planned pairs include MLP versus Transformer on the same profile, embedding on versus off on the same profile, and the 78-dimensional XLSR Transformer with embedding versus XLSR GOP-S direct. PCC is not ranked across unlike feature widths as a single league table.

Multi-seed training (seeds 0 through 4) is applied only to the Kaldi three-dimensional Transformer and the 78-dimensional XLSR-53 Transformer with phone embedding. Validation speakers remain those drawn under seed 0.

RQ4 has no H4. The Kaldi baseline map is frozen within strata. Table 3.13 gives reporting thresholds. Residual taxonomy is applied to XLSR-53 GOP-S and the XLSR-53 78-dimensional Transformer with phone embedding. The residual is $e_i = \hat{y}_i - y_i$. Primary labels use Table 3.14. Class names map to T1 (Alignment Shift), T2 (Phonetic Confusion), T3 (L2 Accent Distortion), T4 (Contextual Coarticulation), and T5 (Speaker Outlier).

Table 3.13: RQ4 strata, locked GOP, and minimum N for reporting PCC/SCC

Stratum	**GOP and map**	**Min N**	**Note**
Phoneme (39 CMU)	Kaldi LPP; baseline map frozen	100 / symbol	Vowel versus consonant is a descriptive grouping
Speaker (125 test)	Same	50 / speaker	Train/test speaker overlap is zero
Sentence-accuracy tertile	Same	50	Speaker-mean accuracy on 0-10; not CEFR

Table 3.14: Locked residual-taxonomy rules

Class	Rule
T2 (Phonetic Confusion)	Human score ≤ 1 and (competitor posterior exceeds canonical, or incorrect/insertion markup)
T3 (L2 Accent Distortion)	Accent markup, or $\lvert y-1\rvert \leq 0.35$ with $\hat{y} \geq 1.5$
T4 (Contextual Coarticulation)	$\lvert e_{\text{self}}\rvert - \lvert e_{\text{other}}\rvert \geq 0.5$ on the two pre-specified systems
T5 (Speaker Outlier)	Speaker in the 0.9 MAE quantile from Behavioral Analysis and $\lvert e\rvert \geq 0.5$
T1 (Alignment Shift)	$\lvert e\rvert \geq 0.75$ and an unusual frame count (≤ 2, ≥ 80, or non-finite)

3.12 Experimental Protocol

This section gathers the operating lock. GOP equations remain in Section 3.7; metric equations remain in Section 3.10; planned Δ pairs remain in Section 3.11. Confirmatory comparisons vary one independent factor at a time, except H3 contrasts against Kaldi, which change both posterior source and GOP formula. Follow-up and learned-scoring rows are not pooled with confirmatory rows.

Table 3.15 is inherited by every phase. Independent factor levels remain those of Table 3.1. Direct maps and OLS use all 125 training speakers. MLP and Transformer scorers use 100 training-fit speakers and 25 validation speakers under seed 0 (Section 3.13). Affine slopes, OLS weights, z-score moments, and network parameters are never fit on test tokens.

Table 3.15: Operating protocol lock for all investigation phases

Component	Locked value	When it may change
Corpus	Speechocean762 only [19]	Never
Official split	125/125 disjoint speakers	Never
Official tokens	Train 47,076; test 47,369	Never (partition counts, not the neural fit set)
Waveform	16 kHz, mono; no augmentation	Never
Scored inventory	39 CMU phones, stress stripped	Never
Join key	Utterance identifier and intra-utterance phone index	Never
Alignment	One Kaldi phone CTM	Never
Human scale	Phoneme 0-2, rounding floor 0.1	Never
Predictions	Clipped to $[0, 2]$	Never
Metrics	PCC, SCC, MAE, MSE; CIs as in Section 3.11	Never
Default seed	0	Multi-seed $\{0, \ldots, 4\}$ only in Section 3.11
Reporting split	Official test only	Train for fitting; validation for early stopping; no test-set model selection

Confirmatory contrasts are H1 on raw PCC versus zero; the six linear H2 pairs; and the three calibrated H3 pairs with GOP-S on SSL. Follow-ups are GOP-SD versus GOP-S within each encoder, and MLP versus Transformer or embedding on versus off on the same feature X.

3.13 Reproducibility

This section states the artifacts, seeds, and checks required to rerun the protocol. Checkpoint identifiers remain those of Table 3.8. Learned hyperparameters remain those of Table 3.9.

Three layers can be reproduced independently: (i) the locked Kaldi 85-dimensional extract for baseline, representation, behavioral, and OLS rows; (ii) frozen SSL inference for GOP-S, GOP-SD, and 78-dimensional profiles; (iii) learned scoring on frozen features with the same 25 validation speakers. Layers (ii) and (iii) do not train an encoder.

Waveforms, scores, and the official split come from Speechocean762 [19]. The phone CTM and 85-dimensional LPP+LPR vectors come from the Kaldi M13 extract and are

not re-decoded on Speechocean762. The 39-phone CMU-to-espeak map is one-to-one; a missing CTC token is a hard failure. Softmax is over the released espeak inventory plus blank ($V = 392$) and is not renormalized onto 39 CMU phones. Every GOP or prediction table joins human scores on `utt_id.phn_idx` with 47,076 training tokens and 47,369 test tokens.

The evaluation stack is Python 3.11 with NumPy, pandas, SciPy, scikit-learn, and PyYAML; learned scoring and SSL forwards use PyTorch and Hugging Face `transformers`. Scientific reproduction is defined as the same Hub identifiers, map, CTM, join keys, metric script, and stored seed-0 predictions. Hardware is not a factor.

Validation speakers are obtained by sorting official training speaker identifiers and drawing $n = \text{round}(125 \times 0.2) = 25$ identifiers without replacement from a NumPy Mersenne Twister initialized at 0. That set is shared across architectures and training seeds. Sanity checks include zero speaker overlap, token-wise identity of standard GOP with canonical LPP, pooled behavioral PCC matching the Kaldi baseline to within 10^{-6}, and headline $\hat{y}$ in $[0,2]$.

This chapter has locked the design, research questions, Speechocean762 extract, pipeline, GOP formulae, scorers, metrics, and inference. Chapter 4 reports the numerical results obtained under that lock.

Chapter 4
Experimental Design and Results

Chapter 3 specified the factorial design, hypotheses H1-H3, and the operating lock. This chapter reports the numerical results. The narrative follows Section 3.2: whether traditional GOP is positively associated with expert scores (RQ1/H1); how representation, acoustic model, and scoring function change that association (RQ2/H2, RQ3/H3, and the learned-scoring extension); and how GOP behaves across phonemes, speakers, and score strata (RQ4, which has no H4).

4.1 Experimental Setup

All tables in this chapter use the official Speechocean762 test partition [19] ($N = 47{,}369$ phoneme tokens; Table 3.4). Acoustic models, GOP formulae, representations, scorers, metrics, and inference rules are those of Sections 3.6-3.13. Confirmatory rows vary one independent factor at a time, except H3 contrasts against Kaldi, which change both the posterior source and the GOP formula and are read as sensitivity analyses. Follow-up and learned-scoring rows are not pooled into a single "better than" claim with the corresponding confirmatory row.

H1 uses raw GOP. H2, H3, learned scoring, MAE, and MSE use train-only maps clipped to $[0, 2]$. Direct maps and OLS are fit on all 125 training speakers; MLP and Transformer scorers use 100 training-fit speakers and 25 validation speakers drawn from the training side only (Section 3.13). The official test set is never used for fitting, early stopping, or model selection.

4.2 GOP Effectiveness

This section answers RQ1/H1: whether traditional GOP (Kaldi M13, canonical LPP, direct scoring) is positively associated with expert phoneme scores when every other factor is locked (Sections 3.7 and 3.12).

On the Kaldi extract, standard GOP coincides with canonical LPP. Headline PCC and SCC for H1 use raw GOP. MAE and MSE use a train-only affine map clipped to $[0,2]$ (Section 3.9); that calibrated space does not enter the rejection of H1. H0 is rejected if the two-sided 95% bootstrap interval of raw PCC lies entirely above zero.

Human scores on the test set were concentrated at the ceiling: 81.3% of tokens had a human score of 2.0 (mean 1.87). Table 4.1 shows that mean canonical LPP rose monotonically across human-score bands, while within-band GOP variation remained large.

Table 4.1: Mean canonical LPP by human-score band on the official test set

Human-score band	N	**Mean GOP**
[0, 0.5)	1,412	5.56
[0.5, 1.5)	1,860	5.89
[1.5, 2]	44,097	6.47

Table 4.2 reports the confirmatory metrics. Raw test PCC was 0.323, with a 95% bootstrap interval of $[0.314, 0.332]$. The interval lay entirely above zero, so H0 of H1 was rejected. The fitted map ($a \approx 0.219$, $b \approx 0.444$) produced a prediction floor of approximately 1.07: no test token mapped near 0. Calibrated PCC (0.342) is not used to reject H1.

Table 4.2: Traditional GOP versus expert phoneme scores. PCC and SCC are computed on raw GOP. MAE and MSE use the train-only affine map clipped to $[0,2]$. Intervals are percentile 95% bootstrap intervals on the test tokens ($n_{\text{boot}} = 1{,}000$).

Metric	**Train**	**Test**	95% **CI (test)**
n	47,076	47,369	-
PCC (raw)	0.342	0.323	[0.314, 0.332]
SCC (raw)	0.329	0.313	[0.304, 0.321]
MAE (mapped)	0.219	0.199	[0.197, 0.202]
MSE (mapped)	0.152	0.120	[0.116, 0.124]

H1 is supported, but the association is modest ($R^2 \approx 0.104$). Scalar Kaldi LPP therefore carries signal and is not a substitute for expert ratings. The next section keeps this extract fixed and varies only representation.

4.3 GOP Representation

This section answers RQ2/H2: how representation changes GOP-human agreement when the Kaldi extract and linear scoring family are locked (Sections 3.8 and 3.9). Scalar LPP and LPR use a univariate affine map; the rank-2 vector and the 84-dimensional profile use OLS. All maps are fit on training tokens and clipped to $[0, 2]$.

On this extract, standard GOP is identical to canonical LPP (an identity check, not an H2 contrast). Table 4.3 reports the five linear configurations.

Table 4.3: Performance comparison across GOP feature representations on the official Speechocean762 test partition ($N = 47{,}369$). All representations are extracted from the Kaldi M13 (TDNN-F Chain) acoustic model and evaluated with train-only linear estimators clipped to $[0, 2]$.

Representation	**Estimator**	**PCC**	**SCC**	**MAE**	**MSE**
Standard GOP	Univariate Affine	0.323	0.313	0.199	0.120
Canonical LPP	Univariate Affine	0.323	0.313	0.199	0.120
Competitor LPR	Univariate Affine	0.328	0.342	0.198	0.121
Rank-2 [LPP, $\max_{q \neq p}$ LPP]	Bivariate OLS	0.351	0.342	0.197	0.120
84-d LPP+LPR Profile	84-d OLS	**0.361**	0.332	**0.197**	**0.119**

H2 is supported: competitor-aware representations raise agreement relative to scalar LPP. LPR improves rank correlation (SCC $= 0.342$ versus 0.313). The rank-2 vector raises PCC to 0.351. The 84-dimensional profile reaches the highest linear PCC (0.361). Expanding dimensionality from 1 to 84 still yields only $\Delta\text{PCC} = +0.038$, which motivates the learned scorers of Section 4.6.

4.4 Acoustic Model Dependency

This section answers RQ3/H3: whether GOP-human agreement depends on the posterior source under direct scoring (Table 3.8; Section 3.7). Encoders remain frozen. Confirmatory SSL GOP is GOP-S; GOP-SD is a graph follow-up and is not pooled into H3.

Table 4.4 reports direct-scoring metrics. Table 4.5 reports the three planned H3 pairs. All three intervals exclude zero, so H3 is supported.

Table 4.4: Acoustic model performance comparison on the official Speechocean762 test set ($N = 47{,}369$). All models use direct scoring with train-only univariate affine maps clipped to $[0, 2]$. Confidence intervals denote percentile 95% bootstrap intervals ($n_{\text{boot}} = 1{,}000$).

Acoustic Model	**GOP Formulation**	**PCC**	**SCC**	**MAE**	**MSE**
Kaldi M13 (TDNN-F Chain)	Canonical LPP	0.323	0.313	0.199	0.120
Wav2Vec2-lv60 (espeak CTC)	Alignment-free GOP-S	0.430	0.334	0.179	0.112
Wav2Vec2-lv60 (espeak CTC)	Alignment-free GOP-SD	0.463	0.346	0.174	0.108
Wav2Vec2-XLSR-53 (espeak CTC)	Alignment-free GOP-S	**0.463**	**0.355**	**0.173**	**0.108**
Wav2Vec2-XLSR-53 (espeak CTC)	Alignment-free GOP-SD	**0.498**	**0.366**	**0.170**	**0.103**

Table 4.5: Planned paired ΔPCC contrasts for hypothesis H3. All intervals exclude zero.

Contrast	ΔPCC	95% **CI**
XLSR-53 GOP-S − Kaldi LPP	+0.121	[+0.111, +0.131]
lv60 GOP-S − Kaldi LPP	+0.088	[+0.077, +0.098]
XLSR-53 GOP-S − lv60 GOP-S	+0.033	[+0.027, +0.040]

Frozen SSL posteriors raise direct GOP-human PCC relative to Kaldi LPP. Multilingual XLSR-53 GOP-S exceeds monolingual lv60 GOP-S by $\Delta\text{PCC} = +0.033$. Switching GOP-S to GOP-SD, with the encoder frozen, adds about +0.035 PCC and raises XLSR-53 to 0.498 with no learnable scoring parameters.

4.5 Behavioral Analysis

This section answers RQ4 by stratifying the locked Kaldi baseline (canonical LPP with the frozen train-only map) across phoneme class, speaker, and sentence-accuracy tertile (Table 3.13). Learned scoring is reported separately in Section 4.6 and is not treated as the confirmatory answer to RQ4.

Table 4.6 reports the strata. Consonants ($n = 28{,}386$) yielded higher PCC and lower MSE than vowels ($n = 18{,}983$). Mean per-speaker PCC was 0.252 ($\sigma = 0.084$; range $[0.078, 0.539]$), below the pooled PCC of 0.323. Among sentence-accuracy tertiles, correlation was highest in the low tertile (PCC = 0.363) and lowest in the high tertile

(PCC = 0.207), while MSE showed the opposite pattern.

Table 4.6: Behavioral analysis of traditional GOP across phoneme classes, speaker distributions, and sentence-accuracy tertiles on the official Speechocean762 test partition ($N = 47{,}369$, 125 speakers). All strata evaluate Kaldi canonical LPP with frozen baseline calibration.

Stratum	Sample Size	PCC	SCC	MAE	MSE
Phoneme Category					
Consonants	$n = 28{,}386$	0.343	0.314	0.184	0.103
Vowels	$n = 18{,}983$	0.307	0.310	0.223	0.147
Speaker Level					
Mean per-speaker	125 speakers	0.252	-	0.200	-
Per-speaker std / range	$\sigma = 0.084$	[0.078, 0.539]	-	-	-
Sentence-Accuracy Tertile					
Low tertile	42 speakers	**0.363**	**0.351**	0.319	0.275
Mid tertile	46 speakers	0.228	0.238	0.157	0.050
High tertile	37 speakers	0.207	0.206	**0.117**	**0.031**

Pooled PCC therefore conceals phonetic, speaker, and stratum structure. Error taxonomy on locked residuals is reported in Section 4.8.

4.6 Learned GOP Scoring

Learned scoring extends RQ1-RQ2: MLP and Transformer maps on frozen features, with or without canonical phone embeddings (Section 3.9; Table 3.9). It is not a separate research question and is not pooled into H1-H3. Table 4.7 lists the eighteen learned configurations.

Table 4.7: Performance of learned scoring architectures across acoustic features and embedding configurations on the official Speechocean762 test partition ($N = 47{,}369$). Predictions are clipped to $[0, 2]$.

Feature Source	Architecture	Embedding	PCC	SCC	MAE	MSE
Kaldi M13 (TDNN-F Chain)						
Compact 3-d Vector	MLP	None	0.362	0.327	0.192	0.118
Compact 3-d Vector	Transformer	None	0.510	0.368	0.158	0.100
84-d LPP+LPR Profile	MLP	None	0.446	0.350	0.168	0.109
84-d LPP+LPR Profile	Transformer	None	0.530	0.379	0.154	0.097
84-d LPP+LPR Profile	MLP	Phone Embed	0.552	0.359	0.159	0.094
84-d LPP+LPR Profile	Transformer	Phone Embed	0.625	0.403	0.148	0.082
Wav2Vec2-lv60 (SSL)						
GOP-S Scalar	MLP	None	0.447	0.334	0.162	0.110
GOP-S Scalar	Transformer	None	0.483	0.319	0.157	0.105
78-d LPP+LPR Profile	MLP	None	0.548	0.365	0.158	0.095
78-d LPP+LPR Profile	Transformer	None	0.624	0.380	0.134	0.083
78-d LPP+LPR Profile	MLP	Phone Embed	0.618	0.379	0.153	0.084
78-d LPP+LPR Profile	Transformer	Phone Embed	0.653	**0.427**	**0.115**	0.079
Wav2Vec2-XLSR-53 (SSL)						
GOP-S Scalar	MLP	None	0.496	0.355	0.167	0.103
GOP-S Scalar	Transformer	None	0.578	0.361	0.154	0.090
78-d LPP+LPR Profile	MLP	None	0.573	0.392	0.142	0.091
78-d LPP+LPR Profile	Transformer	None	0.639	0.415	0.150	0.080
78-d LPP+LPR Profile	MLP	Phone Embed	0.637	0.401	0.136	0.080
78-d LPP+LPR Profile	Transformer	Phone Embed	**0.671**	0.412	0.133	**0.074**

The Transformer outperformed the token-level MLP on matched features. Phone embeddings raised PCC on both Kaldi and SSL profiles. The highest configuration in this study was the XLSR-53 78-dimensional profile with a Transformer and phone embedding (PCC $= 0.671$, MSE $= 0.074$).

4.7 Statistical Analysis

Inferential rules follow Section 3.11. Table 4.8 reports percentile 95% bootstrap intervals for primary configurations. Table 4.9 reports planned paired ΔPCC tests. Table 4.10 reports multi-seed stability for two representative learned models.

Table 4.8: Percentile 95% bootstrap confidence intervals for Pearson correlation ($n_{\text{boot}} = 1{,}000$) on the official Speechocean762 test set ($N = 47{,}369$). All models are evaluated in calibrated $[0, 2]$ score space.

Scoring System	**PCC**	95% **Bootstrap CI**	**SCC**
Kaldi Direct (Calibrated)	0.342	[0.333, 0.351]	0.313
Kaldi Rank-2 Vector (OLS)	0.351	[0.342, 0.361]	0.342
Wav2Vec2-lv60 GOP-S (Direct)	0.430	[0.420, 0.440]	0.334
Kaldi 84-d Profile + MLP	0.446	[0.433, 0.460]	0.350
Wav2Vec2-XLSR-53 GOP-S (Direct)	0.463	[0.453, 0.473]	0.355
Kaldi Compact 3-d + Transformer	0.510	[0.495, 0.523]	0.368
Kaldi 84-d Profile + Transformer	0.530	[0.516, 0.543]	0.379
Wav2Vec2-lv60 78-d + Transformer	0.624	[0.611, 0.637]	0.380
Kaldi 84-d + Transformer + Phone Embed	0.625	[0.612, 0.637]	0.403
Wav2Vec2-XLSR-53 78-d + Transformer	0.639	[0.626, 0.651]	0.415
Wav2Vec2-lv60 78-d + Transformer + Phone Embed	0.653	[0.639, 0.664]	**0.427**
Wav2Vec2-XLSR-53 78-d + Transformer + Phone Embed	**0.671**	[0.660, 0.683]	0.412

Table 4.9: Paired bootstrap difference tests (ΔPCC, $n_{\text{boot}} = 1{,}000$) for planned system comparisons. All confidence intervals exclude zero.

Contrast Pair	ΔPCC	95% **Paired CI**	H_0 **Status**
XLSR-53 GOP-S − Kaldi LPP	+0.121	[+0.111, +0.131]	Rejected
lv60 GOP-S − Kaldi LPP	+0.088	[+0.077, +0.098]	Rejected
XLSR-53 GOP-S − lv60 GOP-S	+0.033	[+0.027, +0.040]	Rejected
Compact 3-d Transformer − Rank-2 OLS	+0.158	[+0.147, +0.168]	Rejected
Kaldi 84-d Transformer − Kaldi 84-d MLP	+0.084	[+0.072, +0.096]	Rejected
Kaldi Transf. + Embed − Kaldi Transf. no Embed	+0.095	[+0.086, +0.104]	Rejected
XLSR-53 Transf. + Embed − XLSR-53 Transf. no Embed	+0.033	[+0.026, +0.039]	Rejected
lv60 Transf. + Embed − lv60 Transf. no Embed	+0.028	[+0.020, +0.038]	Rejected
Full System (XLSR Transf. + Embed) − XLSR GOP-S	+0.208	[+0.198, +0.217]	Rejected

Table 4.10: Multi-seed training stability across seeds $\{0,1,2,3,4\}$ on the official test set.

Architecture	**Mean** PCC $\pm\,\sigma$	**Min** PCC	**Max** PCC
Kaldi Compact 3-d + Transformer	0.509 ± 0.001	0.507	0.510
Wav2Vec2-XLSR-53 78-d Transformer + Embed	0.665 ± 0.004	0.659	0.670

Primary confirmatory intervals excluded zero. Seed-wise standard deviations for the two representative models were small relative to the planned Δ values.

4.8 Error Analysis

Residual taxonomy follows Table 3.14. Two pre-specified systems are compared: Wav2Vec2-XLSR-53 GOP-S with a train-only affine map, and the XLSR-53 78-dimensional Transformer with phone embedding.

Table 4.11: Error taxonomy distribution on the official Speechocean762 test partition ($N = 47{,}369$) comparing the Direct SSL baseline and the Neural Transformer system.

Taxonomy Category	**Direct SSL**	**Neural Transformer**	**Observed Effect**
T3 (L2 Accent Distortion)	4,516	4,545	Dominant error source ($\approx 9.6\%$)
T2 (Phonetic Confusion)	2,149	2,149	Intrinsic acoustic confusion
T4 (Contextual Coarticulation)	402	176	-56.2% reduction
T5 (Speaker Outlier)	202	186	Marginal reduction
T1 (Alignment Shift)	75	108	Small sample fraction ($< 0.25\%$)
Other / Unclassified	40,025	40,205	Accurate / low-residual tokens

T3 remained the largest labeled class. The Transformer reduced T4 counts from 402 to 176 (-56.2%). Table 4.12 shows lower mean absolute error at both score extremes for the Transformer.

Table 4.12: Mean absolute error $|\hat{y} - y|$ at score scale boundaries on the official test set.

Score Region	**Direct SSL Baseline**	**Neural Transformer**
Severe Errors (Human ≈ 0.0)	1.310	**0.910** (-30.5%)
Correct Tokens (Human ≈ 2.0)	0.122	**0.089** (-27.0%)

4.9 Summary of Findings

Table 4.13 ranks the primary scoring configurations on the official test set. Learned scoring is listed as an extension of RQ1-RQ2, not as the confirmatory answer to RQ4.

Table 4.13: Overall performance ranking across primary GOP assessment paradigms on the official Speechocean762 test set ($N = 47{,}369$). All reported configurations use calibrated $[0, 2]$ score spaces.

Rank	Acoustic Model and Scoring Paradigm	PCC	SCC	MAE	MSE
1	**Wav2Vec2-XLSR-53 78-d + Transf. + Phone Embed**	**0.671**	0.412	0.133	**0.074**
2	Wav2Vec2-lv60 78-d + Transf. + Phone Embed	0.653	**0.427**	**0.115**	0.079
3	Wav2Vec2-XLSR-53 78-d + Transf. (no Embed)	0.639	0.415	0.150	0.080
4	Wav2Vec2-XLSR-53 78-d + MLP + Phone Embed	0.637	0.401	0.136	0.080
5	Kaldi 84-d Profile + Transf. + Phone Embed	0.625	0.403	0.148	0.082
-	Wav2Vec2-XLSR-53 Direct GOP-SD (AF-SD)	0.498	0.366	0.170	0.103
-	Wav2Vec2-XLSR-53 Direct GOP-S (AF-S)	0.463	0.355	0.173	0.108
-	Kaldi 84-d Profile (Multivariate OLS)	0.361	0.332	0.197	0.119
-	**Kaldi M13 (TDNN-F Chain Baseline) LPP**	**0.323**	**0.313**	**0.199**	**0.120**

RQ1/H1. Traditional Kaldi LPP is positively associated with expert scores (PCC $= 0.323$, 95% CI $[0.314, 0.332]$). The association is modest and linearly compressed at the bottom of the scale.

RQ2/H2. Competitor-aware representations raise linear agreement; the 84-dimensional profile reaches PCC $= 0.361$.

RQ3/H3. Frozen SSL posteriors exceed Kaldi under direct scoring; XLSR-53 GOP-S yields ΔPCC $= +0.121$ versus Kaldi, and GOP-SD reaches 0.498.

RQ4. Baseline GOP is stronger on consonants than vowels, weaker within speakers than in the pool, and strongest in correlation on the low sentence-accuracy tertile. Residual classes T1-T5 are reported in Table 4.11.

Chapter 5
Discussion

Chapter 4 reported the locked empirical results. This chapter interprets those results: why the associations arise, how they relate to prior work, and what they imply within the declared scope. Headline numbers remain those of Table 4.13; they are not restated as a second results chapter. Learned Transformer scoring is an extension of RQ1-RQ2 and is not treated as the confirmatory answer to RQ4.

5.1 Overview of the Main Findings

Traditional Kaldi GOP is statistically associated with expert ratings but diagnostically limited (RQ1). Competitor-aware representations raise linear fidelity (RQ2). Frozen SSL posteriors raise direct GOP-human PCC, with a further gain from GOP-SD (RQ3). Behavioral strata and the T1-T5 taxonomy show where the baseline fails (RQ4). The highest configuration in this study combines an XLSR-53 confusion profile with a Transformer and phone embedding.

5.2 Discussion of RQ1: Effectiveness of GOP-based Pronunciation Assessment

H1 is supported: canonical LPP tracks expert scores in the correct direction (Table 4.1). That result is consistent with the GOP premise of Witt and Young [16]: higher canonical posterior mass is associated with closer acoustic match to the target phone.

The modest PCC reflects structural limits of scalar Kaldi GOP. Span pooling on a Viterbi CTM mixes adjacent phones when boundaries drift, which is common in L2 durations. Scalar LPP also cannot distinguish diffuse uncertainty from a sharp competitor peak. Linear calibration on a ceiling-heavy score distribution then produces a prediction floor near 1.07, so severe errors remain under-penalized (Table 4.12). Traditional GOP is therefore a valid but incomplete filter, not a standalone grader.

5.3 Discussion of RQ2: Effect of GOP Representation

H2 is supported under a locked Kaldi extract (Table 4.3). LPR raises rank correlation more than linear PCC, which is expected if human ratings track categorical confusion more closely than absolute posterior mass. The rank-2 vector lets the linear map weight canonical evidence and competitor pressure separately, which scalar LPR cannot do. The 84-dimensional profile further encodes which competitors carry mass, matching systematic L1 transfer rather than isotropic noise.

The linear gain from 1 dimension to 84 remains modest ($\Delta\text{PCC} = +0.038$). OLS cannot represent phone-conditioned interactions or utterance context. That plateau motivates the learned scorers of Section 4.6; it does not, by itself, answer RQ4.

5.4 Discussion of RQ3: Effect of Acoustic Model

H3 is supported (Table 4.5): frozen Wav2Vec 2.0 posteriors raise direct GOP-human PCC relative to Kaldi LPP. While these paired gains support H3, the underlying acoustic mechanism stems from convolutional waveform features that retain finer spectral detail than MFCC and from the Transformer context that Wav2Vec 2.0 applies over the full utterance [1]. Contrasts against Kaldi also change the GOP formula (span LPP versus alignment-free GOP-S) and are therefore sensitivity analyses, not matched bake-offs.

XLSR-53 GOP-S exceeds lv60 GOP-S by $\Delta\text{PCC} = +0.033$ even though lv60 is a large English model. A plausible mechanism is that multilingual pretraining [3] represents non-native drift as in-distribution variation rather than collapsing it as out-of-vocabulary noise. Building on this encoder contrast, GOP-SD then adds an explicit deletion penalty and raises both encoders by about $+0.035$ PCC, which aligns with omission as a frequent L2 error class.

Alignment-free GOP-S and GOP-SD evaluate a canonical sequence on CTC posteriors without using Kaldi spans as the scoring window [2]. That design removes one source of boundary leakage for the SSL branch; it does not claim a production latency target.

5.5 Discussion of RQ4: Behavioral Patterns and Failure Modes of GOP

RQ4 concerns the locked Kaldi baseline (Table 4.6). Learned scoring is discussed only where it changes a pre-specified residual class.

Consonants yield higher PCC and lower MSE than vowels. Plosives and fricatives produce localized bursts and frication that concentrate posterior mass, whereas vowels occupy a continuous formant space that L2 speakers often centralize. The drop from pooled PCC (0.323) to mean per-speaker PCC (0.252) indicates that between-speaker accuracy differences inflate the global slope. Within a speaker, unnormalized vocal-tract and rate effects remain.

The tertile pattern is a range-restriction and floor effect, not a CEFR result. High-tertile speakers cluster at score 2.0, which depresses PCC while keeping MSE small. Low-tertile speakers span the scale, so PCC is higher, but the linear floor near 1.07 inflates MSE on tokens rated 0.

T3 (L2 Accent Distortion) is the largest labeled residual class (Table 4.11). The Transformer extension reduces T4 (Contextual Coarticulation) from 402 to 176 tokens, which is consistent with self-attention using neighboring phones. That reduction belongs to the learned-scoring extension, not to the confirmatory RQ4 strata.

5.6 Overall Discussion: What Do the Findings Tell Us About GOP?

Across locked comparisons, the posterior source contributes the largest direct-scoring change, representation supplies competitor geometry that linear maps only partly exploit, and sequence scoring plus phone embeddings account for the remaining gain on frozen profiles (Table 4.13). Scalar Kaldi GOP is therefore limited as a standalone score, but GOP remains useful as an interpretable posterior feature layer for contextual maps. The study does not claim that GOP is obsolete, nor that the hybrid system is a general CAPT product.

5.7 Comparison with Previous Studies

Witt and Young [16] defined GOP from duration-normalized phone posteriors (in practice, a posterior or likelihood ratio against a phone loop). The present Kaldi baseline supports that premise (positive PCC) while showing limits that early work could not stratify at this scale: boundary leakage, missing competitor geometry, and T4 residuals.

Zhang et al. [19] reported phoneme-level PCC of 0.25 for direct GOP and 0.45 for GOP features with per-phone SVRs on Speechocean762. The present Kaldi M13 baseline (PCC $= 0.323$ test) sits between those figures on a comparable order of magnitude. The setups are not identical (a global linear map here rather than per-phone SVRs), so the comparison is a comparable tier rather than a strict replication. The contribution here is the locked factorial and the behavioral strata, including mean per-speaker PCC of 0.252.

Gong et al. [7] reported phone-level correlation near 0.612 for GOPT. The Kaldi 84-dimensional Transformer with phone embedding reaches PCC $= 0.625$, a comparable tier under phone-level scoring only (not the full multi-task GOPT stack). Replacing Kaldi profiles with XLSR-53 78-dimensional profiles raises PCC to 0.671 on this protocol.

Cao et al. [2] introduced alignment-free GOP-S and GOP-SD. The present direct SSL rows reproduce the same direction of gain from GOP-S to GOP-SD on both encoders. The learned extension then maps the 78-dimensional trajectory with a Transformer rather than stopping at affine calibration of a scalar.

5.8 Practical Implications

The study does not implement a user-facing CAPT application. The following points are implications of the locked measurements, not product specifications.

Direct XLSR-53 GOP-SD (PCC $= 0.498$) requires no learnable scoring parameters and does not use Kaldi spans as the scoring window. A hybrid XLSR-53 profile with a Transformer and phone embedding reaches PCC $= 0.671$ on the same test tokens. Which regime is appropriate depends on whether a deployment can host a small sequence scorer; that choice was not field-tested here.

Linear maps on ceiling-heavy labels produce a prediction floor near 1.07 and therefore under-penalize tokens rated 0. Non-linear maps reduced that residual (Table 4.12). The drop from pooled to within-speaker PCC further indicates that speaker-level acoustic offsets remain in raw posteriors; speaker-conditioned calibration is a plausible follow-on, not a result of this protocol.

Competitor slots in the posterior profile identify likely substitutions. Turning those slots into instructional text or a color interface is outside the scope of the study.

5.9 Threats to Validity

Internal Validity. Train and test speakers do not overlap. Every configuration is scored on the same 47,369 test tokens with the same join key and metric script. SSL encoders remain frozen. Neural scorers use a speaker-independent validation split carved from training speakers only (Section 3.12). Kaldi M13 is a single pretrained TDNN-F chain model, so statements about Kaldi GOP apply to that model and not to other Kaldi acoustic models.

External Validity. All headline metrics come from Speechocean762 [19]. Speakers are predominantly Mandarin-L1, so T2 and T3 patterns reflect that transfer. The material is prompted read speech. Cross-corpus, multi-L1, and conversational generalization remain untested.

Construct Validity. Expert phoneme scores on $[0, 2]$ are the construct operationalization. Inter-annotator variance sets an empirical ceiling. GOP is an acoustic posterior proxy, not a biomechanical measurement of articulation.

Statistical Conclusion Validity. H1-H3 use token-level percentile bootstrap intervals and paired ΔPCC (Section 3.11). Primary confirmatory intervals excluded zero in the reported runs. Two representative learned models were repeated across five seeds; seed-wise standard deviations were small relative to the planned Δ values. Token resampling does not remove within-speaker dependence; that dependence is treated as a remaining threat rather than as a change of the analysis unit.

The empirical interpretation in this chapter is taken up as conclusions in Chapter 6.

5.10 Chapter Summary

The discussion supports four claims. Traditional GOP is valid but limited. Competitor geometry helps linear scoring only up to a plateau. Frozen SSL posteriors, especially multilingual XLSR-53 with GOP-SD, raise direct agreement. Behavioral strata and T1-T5 residuals show that pooled PCC is incomplete; sequence scoring reduces T4 as a learned extension of RQ1-RQ2. Chapter 6 restates findings, contributions, and limits.

Chapter 6
Conclusion and Future Work

This chapter restates the study arc, answers RQ1-RQ4, lists contributions and limits, and outlines future work.

6.1 Summary of the Thesis

CAPT systems need phoneme-level evidence that remains interpretable. GOP supplies that evidence from acoustic posteriors [16], but implementations differ in geometry, acoustic model, alignment, and scorer, so reported gains are hard to attribute. This thesis measured those factors on Speechocean762 [19] under one locked protocol (Chapter 3).

Figure 6.1 summarizes the progression: a Kaldi LPP baseline (RQ1), representation variants (RQ2), frozen SSL posteriors (RQ3), and behavioral strata plus residual taxonomy (RQ4), with learned scoring as an extension of RQ1-RQ2.

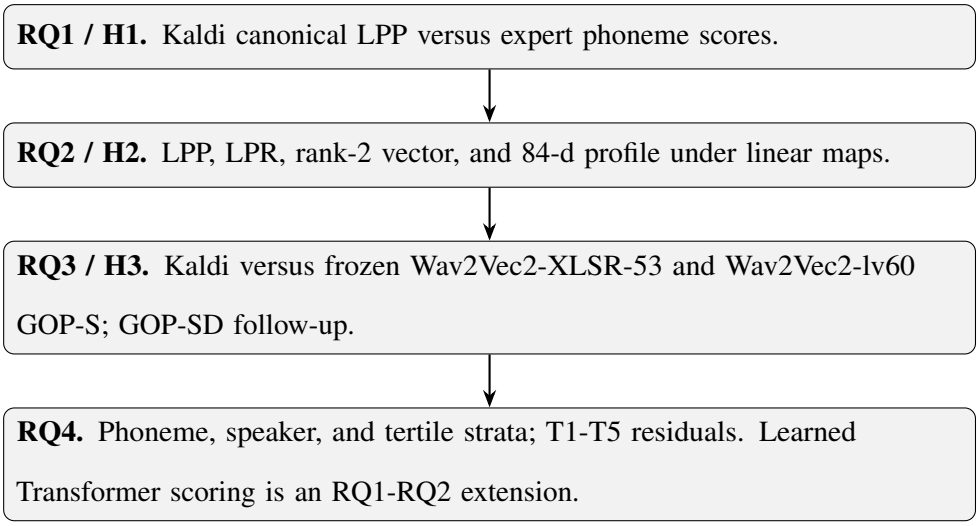


Figure 6.1: Executive research progression of the thesis.

6.2 Key Empirical Findings

Headline metrics are those of Table 4.13.

RQ1/H1. Canonical Kaldi LPP is positively associated with expert phoneme scores (PCC =

0.323, 95% CI $[0.314, 0.332]$). The association explains about 10% of score variance and is linearly compressed at the bottom of the scale.

RQ2/H2. Competitor-aware representations raise linear agreement. LPR improves SCC to 0.342; the rank-2 vector reaches PCC $= 0.351$; the 84-dimensional profile reaches 0.361.

RQ3/H3. Frozen SSL posteriors exceed Kaldi under direct scoring. XLSR-53 GOP-S yields ΔPCC $= +0.121$ versus Kaldi and $+0.033$ versus lv60 GOP-S. GOP-SD raises XLSR-53 to 0.498.

RQ4. Baseline GOP is stronger on consonants than vowels, weaker within speakers (mean PCC 0.252) than in the pool, and shows a tertile pattern driven by range restriction and the linear floor. T3 is the largest labeled residual class. The Transformer extension reduces T4 by 56.2% and is not the confirmatory answer to RQ4.

The highest configuration in this study is the XLSR-53 78-dimensional Transformer with phone embedding (PCC $= 0.671$, MSE $= 0.074$).

6.3 Research Contributions

Theoretical. The results support treating GOP as a posterior feature family rather than a single scalar that deep models should replace. Competitor mass is structured confusion, not unstructured noise.

Methodological. The study specifies a factorial protocol that locks partition, CTM, metrics, and inference while varying one confirmatory factor at a time, with bootstrap intervals and paired ΔPCC (Chapter 3).

Empirical. The locked Speechocean762 evaluation spans the Kaldi baseline, representation variants, frozen SSL GOP-S/GOP-SD, and eighteen learned scoring configurations (Table 4.7). Behavioral strata and the T1-T5 taxonomy document where GOP succeeds and fails under that lock.

Practical implications. Direct GOP-SD and a hybrid Transformer on frozen SSL profiles are two measured scoring regimes. They are implications for system design, not a deployed

CAPT product.

6.4 Limitations of the Study

Headline metrics come from a single corpus, Speechocean762 [19]. Channel, grading norms, and lexical material may differ elsewhere; L2-ARCTIC [20] and related sets were not evaluated.

Speakers are predominantly Mandarin-L1. Confusion patterns therefore reflect Mandarin-English transfer and need not match Romance, Slavic, or other L1 groups.

The material is prompted read speech. Disfluency, repair, and conversational reduction are outside the protocol.

GOP and the learned maps are acoustic-statistical constructs. They are not direct measurements of tongue, lip, or glottal articulation, so any instructional mapping from competitor posteriors remains an inference.

6.5 Directions for Future Work

Cross-corpus and multi-L1 evaluation is the first priority, including multi-accent sets such as L2-ARCTIC [20]. Pediatric speech poses additional F_0 and vocal-tract challenges that this protocol did not test.

Segmental GOP does not score stress, rhythm, or intonation. Joint models that add prosodic features remain future work.

The within-speaker PCC drop motivates lightweight speaker adaptation of frozen SSL features after a short calibration utterance. Such adaptation was not run here.

Phoneme-level highlighting can surface errors without generating explanatory dialogue [4]. Whether competitor posteriors can condition a language-model tutor is an open interface question; this thesis does not evaluate that coupling.

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